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Dark-Knowledge-Guided Surrogate Distillation for Query-Efficient Black-Box Adversarial Attacks on Remote Sensing Image Classification

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***Abstract*—Deep neural networks (DNNs) have achieved remarkable performance in remote sensing image (RSI) scene classification, yet they remain highly vulnerable to adversarial examples, raising serious security concerns for automated Earth observation systems. In realistic black-box settings, the target model is hidden behind an inference interface and can only be queried for output probabilities, which makes adversarial attacks challenging. Query-based attacks require a large number of queries, whereas transfer-based attacks suffer from unstable transferability when the surrogate model poorly approximates the target. To bridge this gap, this article proposes a dark-knowledge-guided surrogate distillation framework for query-efficient black-box adversarial attacks on RSI classification. The key insight is that the value of a query lies not in its quantity but in its information density: a query is more informative when the target model's response reveals richer inter-class relations. Motivated by this, we design RsMix, a dark-knowledge-guided semantic mixing strategy that progressively estimates a category confusion matrix from the target model's top-k responses and biases the mixing toward semantically similar RSI categories, thereby producing high-dark-knowledge samples that expose richer class relations under a limited query budget. A top-k dark-knowledge distillation objective is then formulated to transfer the most informative target-model responses to the surrogate, and a classification feature alignment loss is introduced to organize the surrogate feature space consistently with the target model's top-k class relations without accessing target-model internals. Finally, adversarial examples are generated on the distilled surrogate under a manifold consistency constraint that preserves the semantic naturalness of RSI scenes. Extensive experiments on the AID, NWPU-RESISC45 and MLRSNet datasets across multiple surrogate-target combinations, including convolutional and transformer architectures, demonstrate that the proposed method achieves higher attack success rates and stronger cross-model transferability than representative baselines while maintaining query efficiency and visual imperceptibility. These results reveal the security risks of RSI interpretation systems under query-efficient black-box attacks and provide insights for designing more robust remote sensing models.**

***Index Terms*—Adversarial examples, remote sensing image classification, black-box attack, query efficiency, knowledge distillation, dark knowledge, surrogate model, transferability.**

I. Introduction

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DEEP neural networks (DNNs) have achieved remarkable performance in remote sensing image (RSI) interpretation tasks, including scene classification [1]–[4], object detection [5], semantic segmentation [6], change detection [7], and land-cover mapping [8]. Benefiting from large-scale representation learning and advanced network architectures, modern RSI classifiers can extract discriminative spatial, spectral, and semantic features from complex Earth observation data. These capabilities have enabled a wide range of safety-critical applications, such as military reconnaissance [9], disaster assessment [10], urban planning [11], environmental monitoring [12], and precision agriculture [13]. Despite these achievements, recent studies have shown that DNN-based RSI interpretation systems remain vulnerable to adversarial examples [14]. By adding carefully crafted but visually imperceptible perturbations to clean images, adversaries can mislead DNNs into producing incorrect predictions with high confidence [15].

The vulnerability of RSI classifiers raises serious security concerns. Remote sensing images often contain mission-critical information related to airports, ports, bridges, residential regions, storage tanks, and transportation infrastructure. If an automated RSI interpretation system is deceived by adversarial perturbations, incorrect scene recognition may lead to unreliable downstream decisions. To intuitively illustrate this effect, Fig. 1 visualizes the attention responses of different RSI classifiers before and after adversarial perturbation. The clean image leads models to focus on discriminative scene structures, whereas the adversarial image shifts high-attention regions toward irrelevant or background areas, causing incorrect predictions. This attention deviation indicates that adversarial perturbations can alter the model's feature utilization while remaining visually subtle.

This risk becomes more severe in black-box scenarios, where the attacker does not know the target model's architecture, parameters, or gradients. In practical remote sensing services, deployed models are usually hidden behind inference interfaces, and only output labels or confidence scores can be queried. Therefore, black-box adversarial attacks provide a realistic way to evaluate the security of remote sensing recognition systems.

Existing black-box attacks can be mainly divided into query-based attacks and transfer-based attacks. Query-based attacks estimate gradient directions or decision boundaries by repeatedly probing the target model [16], [17]. Although these methods can achieve high attack success rates, they

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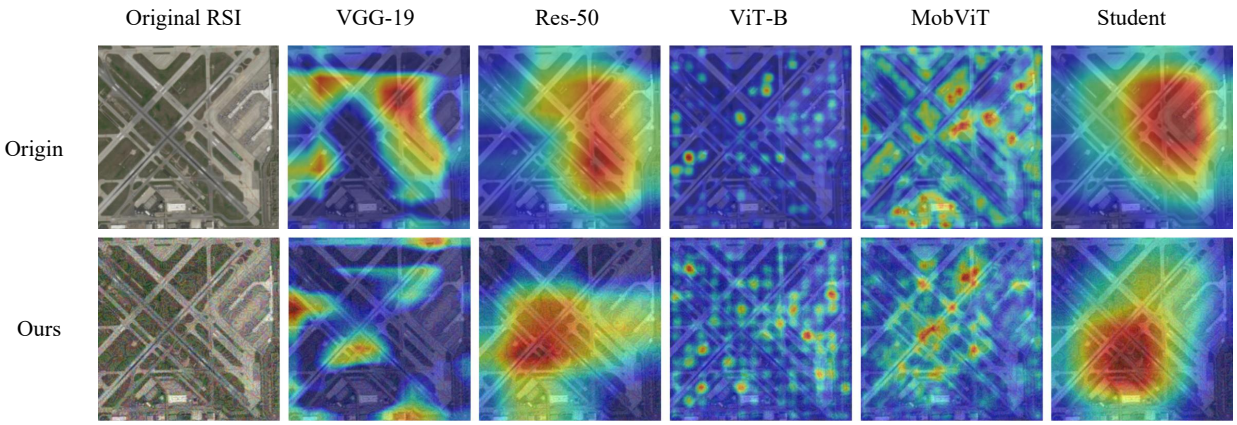

Fig. 1. Attention visualization of different remote sensing image classifiers before and after the proposed attack. Grad-CAM is used for CNN-based models, while gradient-based saliency maps are used for transformer and hybrid architectures. Warmer colors indicate regions with stronger attention. Compared with the clean image, the adversarial example generated by the proposed method shifts model attention from discriminative scene structures to irrelevant regions, leading to misclassification while preserving visual similarity.

usually require a large number of queries, especially for high-resolution remote sensing images with complex spatial structures. Excessive queries increase the attack cost and make the attack easier to detect in practical systems. Transfer-based attacks generate adversarial examples on a source model and directly apply them to the target model [18], [19]. These methods require few or no target-model queries, but their performance heavily depends on the similarity between the source and target decision boundaries. When the source model poorly approximates the target model, adversarial examples may overfit the source model and fail to transfer.

To bridge the gap between query-based and transfer-based attacks, surrogate-model distillation has emerged as a promising direction for black-box attacks. The attacker queries the target model on selected samples and uses the returned soft labels to train a surrogate model. Adversarial examples are then generated on the surrogate and transferred to the target model. However, directly applying conventional distillation to remote sensing black-box attacks faces two limitations. First, clean remote sensing images often produce over-confident softmax outputs, where most probability mass is assigned to the top-1 class. Such peaked responses contain limited dark knowledge about inter-class relationships and provide weak supervision for surrogate learning. Second, remote sensing categories often exhibit strong visual and semantic similarity. For example, port, river, and bridge scenes share water and linear structures; dense residential, medium residential, and commercial areas share building layouts; industrial areas and storage tanks share man-made textures. Conventional distillation does not explicitly exploit these remote-sensing-specific category relations.

In this article, we argue that the key to query-efficient black-box attacks is not merely reducing the number of queries, but increasing the information density of each query. A query is more valuable when the target model's response reveals richer class relations and decision-boundary clues. In knowledge distillation, such non-top-1 probabilities are referred to as dark knowledge. For RSI classification, dark knowledge is particularly meaningful because visually similar scene categories often occupy nearby regions in the target model's feature space. If a queried sample can activate multiple related categories in the target model's top-k response, the returned soft label provides stronger supervision for surrogate distillation than a highly peaked clean-image response.

Motivated by this observation, this article proposes a dark-knowledge-guided surrogate distillation framework for query-efficient black-box adversarial attacks on remote sensing images. The proposed framework first constructs informative mixed remote sensing samples through RsMix, a dark-knowledge-guided semantic mixing strategy. RsMix begins with random class-pair exploration and queries the black-box target model to estimate which category pairs produce richer top-k responses. A category confusion matrix is then progressively updated according to the dark-knowledge gain of each mixed category pair. In later iterations, this matrix guides the generation of mixed samples, making semantically related and visually similar categories more likely to be combined. In this way, RsMix produces high-dark-knowledge samples that reveal richer target-model class relations under a limited query budget.

To train a stronger surrogate model, we formulate a top-k dark-knowledge distillation objective that focuses on the most informative categories in the target model's response. Instead of forcing the surrogate to match the entire softmax distribution, which may include noisy low-probability classes, the surrogate is trained to match the normalized top-k probabilities returned by the target model. This objective allows the surrogate to learn target-model confusion relationships among remote sensing categories. In addition, we introduce a classification feature alignment loss. Since the target model is black-box, its internal features cannot be accessed. We therefore use the target model's top-k probabilities as semantic weights to construct a target-induced class prototype in the surrogate classifier space. By aligning surrogate features with this prototype, the surrogate learns not only target-model outputs but also a feature organization consistent with the target model's top-k class relations.

After surrogate distillation, adversarial examples are gener-

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ated on the distilled surrogate and evaluated on the black-box target model. To preserve the semantic naturalness of remote sensing images, we introduce a manifold consistency constraint during adversarial generation. This constraint keeps adversarial examples close to the corresponding clean images in a semantic embedding space, reducing the risk of disrupting important remote sensing structures such as runways, roads, ports, and residential layouts. The final attack is conducted under an L_∞ perturbation budget, ensuring pixel-level imperceptibility while improving cross-model transferability.

The proposed framework integrates four components: dark-knowledge-enhanced surrogate distillation, RsMix semantic mixing, classification feature alignment, and manifold-constrained adversarial generation. These components jointly address the central difficulty of query-efficient black-box attacks: how to obtain a target-consistent surrogate model with limited target-model feedback and then generate transferable adversarial examples from it. Unlike pure query-based methods, our method avoids repeatedly optimizing perturbations through target-model queries. Unlike conventional transfer attacks, our method improves surrogate fidelity by exploiting the target model's top-k dark knowledge. Unlike generic data augmentation, RsMix explicitly uses target-model feedback to identify category pairs that expose meaningful remote sensing class relations.

The main contributions of this article are summarized as follows:

1) A dark-knowledge-enhanced surrogate distillation framework is proposed for query-efficient black-box adversarial attacks on remote sensing images. The framework uses limited target-model softmax feedback to train a surrogate model that captures top-k class relations and target-model decision behavior.
2) A dark-knowledge-guided remote sensing semantic mixing strategy, termed RsMix, is designed to generate high-information query samples. RsMix progressively estimates a category confusion matrix according to top-k dark-knowledge gain and uses it to guide mixed-sample generation, increasing the information density of each target-model query.
3) A classification feature alignment loss is introduced to improve surrogate learning under the black-box setting. The loss constructs target-induced class prototypes in the surrogate classifier space using target-model top-k probabilities, enabling the surrogate feature representation to better reflect target-model class relations without accessing target-model internals.
4) A manifold-constrained adversarial generation objective is formulated to preserve remote sensing semantic structure during attack generation. The constraint keeps adversarial examples close to clean images in a semantic embedding space while maintaining the L_∞ perturbation budget.

II. Problem Description

A. Problem Formulation

Given a remote sensing image $x \in \mathbb{R}^{H\times W\times 3}$ and its corresponding ground-truth label $y \in \{1, 2, \ldots, C\}$, a remote sensing scene classifier $f : \mathbb{R}^{H\times W\times 3} \to \mathbb{R}^C$ predicts the scene category by

$$\hat{y} = \arg\max_c f_c(x), \tag{1}$$

where $f_c(x)$ denotes the logit or confidence score of the c-th class. The objective of adversarial attacks is to generate an adversarial example x_{adv} that misleads the classifier while preserving the visual and semantic content of the original remote sensing image:

$$\arg\max_c f_c(x_{\text{adv}}) \neq y, \tag{2}$$

subject to a perturbation constraint

$$x_{\text{adv}} - x_\infty \leq \epsilon. \tag{3}$$

In this work, we focus on query-efficient black-box adversarial attacks for remote sensing image classification. The attacker has no access to the internal architecture, parameters, intermediate features, or gradients of the target model. The target model can only be queried to obtain output probabilities or top-k softmax responses. This setting is more realistic for deployed remote sensing interpretation systems, where commercial or operational models are usually hidden behind an inference interface.

Let f_T denote the black-box target model and f_S denote a source model available to the attacker. For an input image x, the attacker can query the target model and obtain

$$p_T(x) = f_T(x), \tag{4}$$

where $p_T(x) \in \mathbb{R}^C$ is the output probability distribution over C scene classes. During adversarial example generation, gradients from f_T are unavailable. Therefore, a surrogate model f_S^* is constructed to approximate the target model's decision behavior. Adversarial examples are generated on f_S^* and then transferred to f_T for black-box evaluation.

The key challenge is how to construct a high-quality surrogate model with a limited number of target-model queries. Directly querying clean remote sensing images often produces over-confident softmax responses, where most probability mass concentrates on the top-1 category. Such responses provide limited information about inter-class similarity and decision-boundary structure. To overcome this limitation, our method constructs high-dark-knowledge mixed samples and uses the target model's top-k responses to distill a more informative surrogate model.

B. Black-Box Query and Dark Knowledge

In knowledge distillation, the non-maximum entries of a softmax vector contain valuable information about class relationships. This information is commonly referred to as dark knowledge. In remote sensing scene classification, dark knowledge is particularly important because many categories exhibit strong visual and semantic similarity. For instance, port, river, and bridge scenes may share water and linear structures; dense residential, medium residential, and commercial areas may share building layouts; industrial areas and storage tanks often contain similar man-made textures.

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For a queried image x, the target model returns a probability distribution

$$p_T(x) = [p_T(1|x), p_T(2|x), \ldots, p_T(C|x)]. \tag{5}$$

We define the top-k class set as

$$\Omega_k(x) = (p_T(x)). \tag{6}$$

The normalized top-k probability is computed as

$$\hat{p}_T(c|x) = \frac{p_T(c|x)}{\sum_{j\in\Omega_k(x)} p_T(j|x)}, \quad c \in \Omega_k(x). \tag{7}$$

Compared with hard labels, the top-k soft response reveals richer category relations. A hard label only indicates the most likely category, while the top-k distribution describes how the target model distributes confidence among semantically related categories. This information can be used to guide surrogate learning under a limited query budget.

However, clean samples often produce highly peaked distributions. To increase the information density of each query, we introduce mixed remote sensing samples. These samples are designed to activate multiple related categories in the target model, thereby producing richer top-k responses. The resulting dark knowledge is used to supervise surrogate distillation.

C. Query-Efficient Surrogate Distillation Objective

The proposed method aims to learn a surrogate model f_S^* that approximates the target model f_T using limited query feedback. The surrogate is trained using three complementary supervision signals: clean-image dark-knowledge distillation, RsMix dark-knowledge distillation, and classification feature alignment.

For a clean image x, the clean dark-knowledge distillation loss is defined as

$$\mathcal{L}_{KD}^{\text{clean}} = -\sum_{c\in\Omega_k(x)} \hat{p}_T(c|x) \log \hat{p}_S(c|x), \tag{8}$$

where $\hat{p}_S(c|x)$ denotes the normalized top-k probability predicted by the surrogate model over the target model's top-k class set.

For an RsMix sample $\tilde{x}_{ij}$ generated from two remote sensing images x_i and x_j, the target model returns $p_T(\tilde{x}_{ij})$. The corresponding mixed-sample dark-knowledge distillation loss is

$$\mathcal{L}_{KD}^{\text{mix}} = -\sum_{c\in\Omega_k(\tilde{x}_{ij})} \hat{p}_T(c|\tilde{x}_{ij}) \log \hat{p}_S(c|\tilde{x}_{ij}). \tag{9}$$

This term encourages the surrogate model to learn the class confusion relations exposed by high-dark-knowledge mixed samples. Since RsMix samples are constructed to elicit richer top-k responses, $\mathcal{L}_{KD}^{\text{mix}}$ provides more informative supervision than ordinary clean-image distillation.

When ground-truth labels are available, a standard classification loss is also introduced to preserve the surrogate model's basic scene discrimination capability:

$$\mathcal{L}_{CE} = -\log p_S(y|x). \tag{10}$$

In addition, the classification feature alignment loss $\mathcal{L}_{cf}$ constrains the surrogate feature representation to approach a target-induced class prototype in the surrogate classifier space. This term helps the surrogate learn not only the target model's output probabilities but also a feature organization consistent with the target model's top-k class relations.

The total surrogate distillation objective is formulated as

$$\mathcal{L}_{\text{sur}} = \lambda_1 \mathcal{L}_{KD}^{\text{clean}} + \lambda_2 \mathcal{L}_{KD}^{\text{mix}} + \lambda_3 \mathcal{L}_{cf} + \lambda_4 \mathcal{L}_{CE}, \tag{11}$$

where λ_1, λ_2, λ_3, and λ_4 are balancing coefficients. The clean distillation term preserves the target model's response on natural remote sensing images. The mixed-sample distillation term transfers richer category confusion knowledge. The classification feature alignment term improves surrogate feature geometry. The cross-entropy term prevents the surrogate from drifting away from the original data distribution.

Through this objective, the surrogate model learns both sample-level prediction behavior and category-level confusion structure from the target model. This improves the reliability of surrogate gradients and enhances black-box transferability.

D. Adversarial Example Generation

After the surrogate model is distilled, adversarial examples are generated on f_S^* and transferred to the target model f_T. For an untargeted attack, the adversarial objective is to increase the classification loss of the surrogate model with respect to the ground-truth label:

$$\mathcal{L}_{\text{atk}} = \mathcal{L}_{CE}(f_S^*(x_{\text{adv}}), y). \tag{12}$$

The adversarial example is generated by solving

$$\max_{x_{\text{adv}}} \quad \mathcal{L}_{\text{atk}}, \tag{13}$$

subject to

$$x_{\text{adv}} - x_\infty \le \epsilon, \quad x_{\text{adv}} \in [0,1]^{H\times W\times 3}. \tag{14}$$

To preserve the semantic naturalness of remote sensing images, we further introduce a manifold consistency constraint. Let $\phi(\cdot)$ denote a semantic feature encoder. The manifold consistency loss is

$$\mathcal{L}_{\text{mc}} = \phi(x_{\text{adv}}) - \phi(x)_2^2. \tag{15}$$

The final adversarial optimization objective becomes

$$\max_{x_{\text{adv}}} \quad \mathcal{L}_{\text{atk}} - \alpha \mathcal{L}_{\text{mc}}, \tag{16}$$

subject to the same ℓ_∞ constraint. The coefficient α balances attack strength and semantic consistency. The perturbation budget ϵ restricts the maximum per-pixel modification, ensuring visual imperceptibility under the ℓ_∞ norm.

For iterative attacks, the update rule can be written as

$$x_{\text{adv}}^{t+1} = \Pi_\epsilon \left(x_{\text{adv}}^t + \eta \cdot \text{sign}\left(\nabla_{x_{\text{adv}}^t} [\mathcal{L}_{\text{atk}} - \alpha \mathcal{L}_{\text{mc}}] \right) \right), \tag{17}$$

where η is the step size and Π_ϵ projects the adversarial image back into the feasible ℓ_∞ ball centered at the clean image x. For single-step FGSM, the update is

$$x_{\text{adv}} = \Pi_\epsilon \left(x + \epsilon \cdot \text{sign}\left(\nabla_x \mathcal{L}_{\text{atk}} \right) \right). \tag{18}$$

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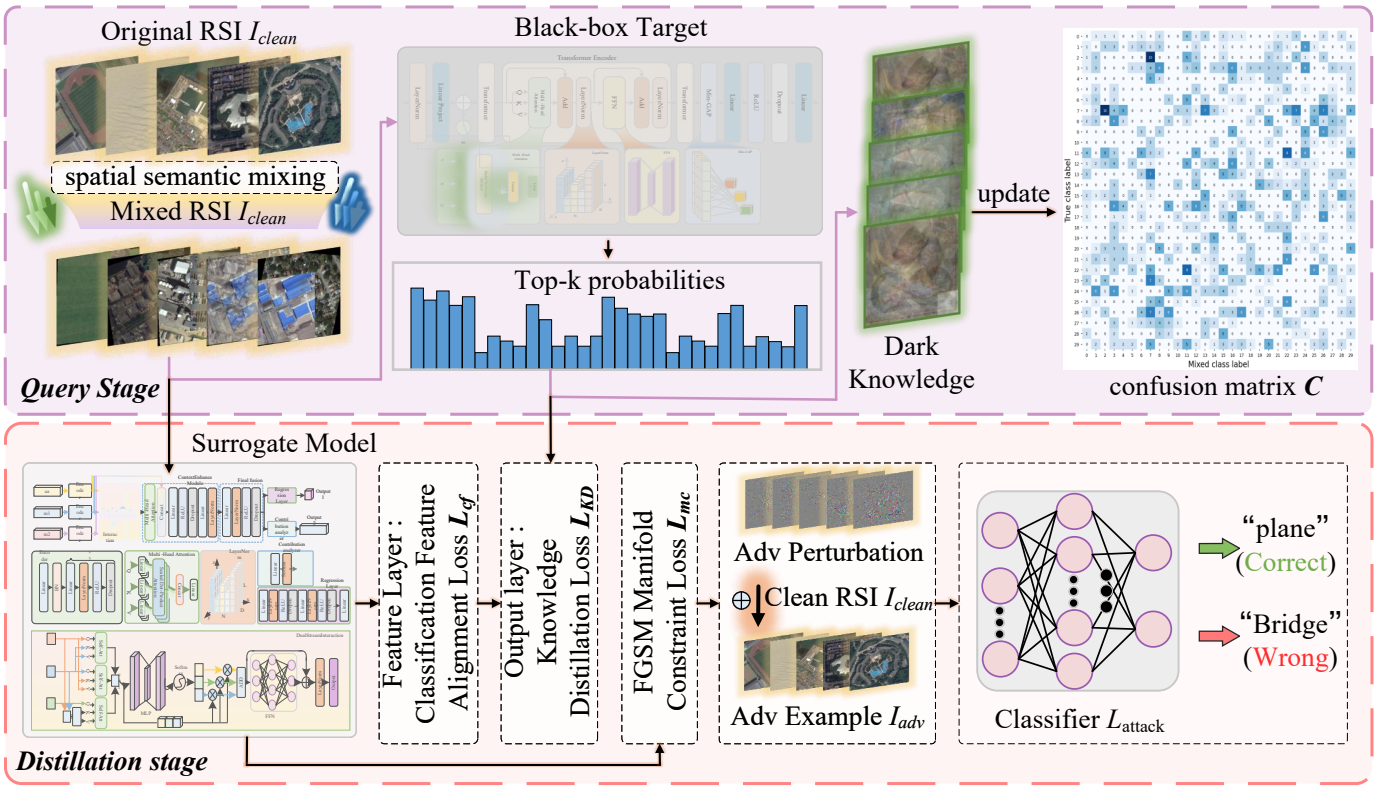

Fig. 2. Overview of the proposed dark-knowledge-guided surrogate distillation framework. The framework contains two stages: query and distillation. In the query stage, original remote sensing images are converted into mixed samples through spatial semantic mixing and submitted to the black-box target model. The returned top-k probabilities provide dark knowledge for updating the category confusion matrix (C), which guides subsequent RsMix sample generation. In the distillation stage, the surrogate model is optimized by the knowledge distillation loss ($\mathcal{L}_{KD}$), classification feature alignment loss ($\mathcal{L}_{cf}$), and manifold constraint loss ($\mathcal{L}_{mc}$). Adversarial examples are then generated on the distilled surrogate to improve black-box transferability while preserving remote sensing semantic structure.

The generated adversarial examples are finally evaluated by querying the black-box target model. An attack is considered successful when

$$\arg\max_{c} f_T^c(x_{\text{adv}}) \neq y. \tag{19}$$

III. Methodology

Fig. 2 presents the overall architecture of the proposed dark-knowledge-guided surrogate distillation framework for query-efficient black-box adversarial attacks on remote sensing images. The entire process can be summarized as follows. The method begins with a source model trained on the same remote sensing dataset. In the early stage, category pairs are randomly sampled to generate initial RsMix samples. The target model is then queried with these samples, and the resulting top-k responses are used to estimate the dark-knowledge gain of each category pair. The estimated gain updates the category confusion matrix, which guides later RsMix sampling. Through this iterative process, the surrogate model receives richer supervision from fewer queries. Finally, adversarial examples generated on the distilled surrogate are transferred to the black-box target model for evaluation.

A. Dark Knowledge in Black-Box Surrogate Distillation

Knowledge distillation usually transfers information from a teacher model to a student model by matching their softened output distributions. In the black-box attack setting, this idea is particularly useful because the target model cannot be directly accessed except through model queries. However, directly querying clean remote sensing images often provides limited supervision. For a correctly classified image, the target model may output a highly confident distribution such as

$$p_T(x) = [0.98, 0.01, 0.004, \ldots], \tag{20}$$

where only the top-1 class receives a meaningful probability. Such a response reveals little about class similarity or decision-boundary structure. In contrast, a response such as

$$p_T(\tilde{x}) = [0.45, 0.25, 0.12, 0.08, \ldots] \tag{21}$$

contains richer dark knowledge, since several categories receive non-negligible probabilities. In remote sensing image classification, this information is especially important because many categories are visually and semantically close, such as airport and runway, port and river, dense residential and medium residential, or industrial area and storage tanks.

We define the top-k class set of the target model for a sample x as

$$\Omega_k(x) = (p_T(x)). \tag{22}$$

The normalized top-k probability is computed as

$$\hat{p}_T(c|x) = \frac{p_T(c|x)}{\sum_{j\in\Omega_k(x)} p_T(j|x)}, \quad c \in \Omega_k(x). \tag{23}$$

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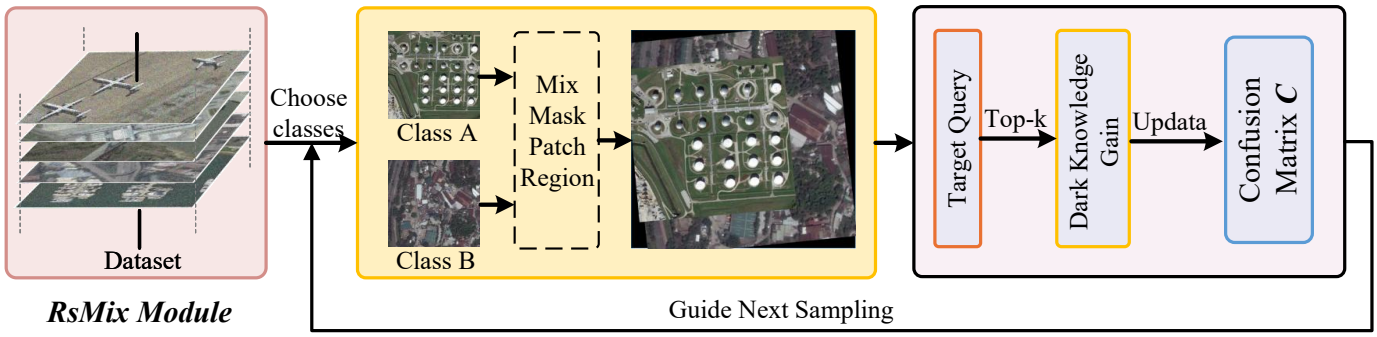

Fig. 3. RsMix module for dark-knowledge-guided remote sensing semantic mixing. Two remote sensing images are first selected according to the category-pair sampling strategy and combined by a spatial mixing mask to generate a mixed sample. The mixed sample is then queried on the black-box target model to obtain top-k probabilities. The resulting dark-knowledge gain updates the category confusion matrix, which guides subsequent sample mixing toward more informative semantic category pairs.

Instead of forcing the surrogate to match the entire softmax distribution, which may contain noisy low-probability classes, we focus on the top-k response:

$$\mathcal{L}_{KD} = -\sum_{c\in\Omega_k(x)} \hat{p}_T(c|x)\log \hat{p}_S(c|x), \tag{24}$$

where $\hat{p}_S(c|x)$ is the surrogate model's normalized probability over the same top-k class set. This top-k dark-knowledge distillation objective encourages the surrogate model to capture the target model's most informative inter-class relationships.

For remote sensing scenes, such top-k relations provide implicit clues about semantic proximity. For example, if a port image receives high probabilities for port, river, bridge, and industrial area, the target model is revealing how these scene categories are arranged near each other in its decision space. Learning such relations allows the surrogate model to better approximate the target model's local decision behavior, thereby improving transfer attack performance.

B. RsMix: Dark-Knowledge-Guided Remote Sensing Semantic Mixing

To increase the amount of dark knowledge obtained from each query, we propose RsMix, a remote sensing semantic mixing strategy guided by target-model feedback. The purpose of RsMix is not merely to augment data, but to generate query samples that can activate multiple semantically related categories in the target model's softmax output. As illustrated in Fig. 3, RsMix first constructs mixed samples from category pairs, then uses the target model's top-k responses to update the category confusion matrix and guide subsequent mixing. These high-dark-knowledge samples allow the surrogate model to learn richer class relations under a limited query budget.

Given two remote sensing images x_i and x_j with labels y_i and y_j, RsMix generates a mixed image $\tilde{x}_{ij}$ as

$$\tilde{x}_{ij} = M \odot x_i + (1-M)\odot x_j, \tag{25}$$

where $M \in \{0,1\}^{H\times W}$ is a spatial mixing mask and $\odot$ denotes element-wise multiplication. The mask can correspond to a rectangular region or a spatially localized area. The effective mixing ratio is defined as

$$\lambda_{ij} = \frac{1}{HW}\sum_{u=1}^{H}\sum_{v=1}^{W} M(u,v). \tag{26}$$

Thus, λ_{ij} indicates the proportion of pixels inherited from x_i, while $1-\lambda_{ij}$ indicates the proportion inherited from x_j.

A naive random mixing strategy may combine unrelated categories, such as desert and baseball field, producing samples that are visually unnatural and weakly informative for distillation. RsMix instead learns which category pairs are likely to produce high-value top-k responses. To achieve this, we maintain a category confusion matrix

$$C \in \mathbb{R}^{C\times C}, \tag{27}$$

where C_{ij} measures how effectively the category pair (i,j) elicits dark knowledge from the target model.

In the initial stage, category pairs are sampled randomly to explore the target model's response behavior. For each mixed sample $\tilde{x}_{ij}$, the target model returns

$$p_T(\tilde{x}_{ij}) = f_T(\tilde{x}_{ij}). \tag{28}$$

We then compute a dark-knowledge gain score. One practical definition is the probability mass assigned to non-source classes within the top-k set:

$$G_{ij} = \sum_{c\in\Omega_k(\tilde{x}_{ij}),\ c\neq y_i,\ c\neq y_j} p_T(c|\tilde{x}_{ij}). \tag{29}$$

A larger G_{ij} indicates that the mixed sample activates not only the two source categories but also other semantically related classes. This suggests that the sample contains richer dark knowledge and is more useful for surrogate distillation.

The confusion matrix is updated by exponential moving average:

$$C_{ij} \leftarrow \beta C_{ij} + (1-\beta)G_{ij}, \tag{30}$$

where $\beta \in [0,1)$ is a smoothing coefficient. Since category confusion is usually symmetric in remote sensing scenes, we set

$$C_{ji} = C_{ij}. \tag{31}$$

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After the exploration stage, category pairs are sampled according to the learned confusion matrix:

$$P(i,j) = \frac{\exp(C_{ij}/\tau)}{\sum_{m=1}^{C}\sum_{n=1}^{C}\exp(C_{mn}/\tau)}, \tag{32}$$

where τ is a temperature parameter controlling the sharpness of the sampling distribution. Category pairs with higher dark-knowledge gain are selected more frequently. This process biases RsMix toward semantically close or visually similar categories, such as port–river, storage tanks–industrial area, or residential–commercial scenes.

The target response of an RsMix sample can be used directly as teacher supervision:

$$p_T(\tilde{x}_{ij}) = f_T(\tilde{x}_{ij}).$$

The surrogate model is then optimized by matching this response:

$$\mathcal{L}_{\text{RsMix}} = -\sum_{c\in\Omega_k(\tilde{x}_{ij})} \hat{p}_T(c|\tilde{x}_{ij}) \log \hat{p}_S(c|\tilde{x}_{ij}). \tag{33}$$

Compared with clean-sample distillation, RsMix increases the information density of each query. A single mixed sample may reveal multiple class relations that are not visible from clean images, allowing the surrogate model to learn a more target-consistent decision structure with fewer target-model queries.

C. Classification Feature Alignment Loss

Although top-k dark-knowledge distillation aligns the output distributions of the target and surrogate models, output matching alone does not guarantee that the surrogate's internal feature space captures the target model's class relation structure. The surrogate may produce similar softmax probabilities while relying on unstable or poorly organized features. Such a mismatch can reduce adversarial transferability because adversarial perturbations are generated from the surrogate model's gradients, which depend heavily on its feature representation.

To address this issue, we introduce a classification feature alignment loss $\mathcal{L}_{cf}$. Since the target model is black-box, its intermediate features and classifier weights are unavailable. Therefore, direct feature matching between the target and surrogate models is impossible. Instead, we use the target model's top-k probabilities as semantic weights to construct a class prototype in the surrogate classifier space.

Let $z_S(x) \in^d$ denote the feature vector extracted by the surrogate model before its final classification layer. Let

$$W_S = [w_1^S, w_2^S, \ldots, w_C^S] \in^{C\times d} \tag{34}$$

denote the surrogate model's classifier weights, where w_c^S is the class direction for category c. Given the target model's top-k response, we construct a target-induced class prototype in the surrogate feature space:

$$q_S(x) = \sum_{c\in\Omega_k(x)} \hat{p}_T(c|x) w_c^S. \tag{35}$$

This prototype represents the class direction implied by the target model's soft label, but expressed within the surrogate model's own classifier space. The classification feature alignment loss is defined as

$$\mathcal{L}_{cf} = 1 - \frac{z_S(x)^\top q_S(x)}{\|z_S(x)\|_2 \|q_S(x)\|_2}. \tag{36}$$

Minimizing $\mathcal{L}_{cf}$ encourages the surrogate feature $z_S(x)$ to approach the target-induced class prototype $q_S(x)$. As a result, the surrogate model learns not only the target model's output probabilities but also a feature-space organization consistent with the target's top-k class relations.

For an RsMix sample $\tilde{x}_{ij}$, this loss becomes

$$\mathcal{L}_{cf}^{mix} = 1 - \frac{z_S(\tilde{x}_{ij})^\top q_S(\tilde{x}_{ij})}{\|z_S(\tilde{x}_{ij})\|_2 \|q_S(\tilde{x}_{ij})\|_2}. \tag{37}$$

This design preserves the black-box assumption because it does not access target-model gradients or intermediate activations. It only uses the target's output probabilities. In practice, $\mathcal{L}_{cf}$ provides a direct feature-level training signal, accelerates surrogate distillation, and improves the transferability of adversarial examples generated from the surrogate.

D. Manifold-Constrained Adversarial Example Generation

After obtaining the distilled surrogate model, adversarial examples are generated on the surrogate and then transferred to the black-box target model. Given a clean image x with label y, the adversarial example is defined as

$$x_{adv} = x + \delta, \tag{38}$$

subject to the L_∞ perturbation constraint

$$\|\delta\|_\infty \le \epsilon. \tag{39}$$

The attack objective aims to mislead the surrogate model. For untargeted attacks, the classification loss can be written as

$$\mathcal{L}_{atk} = \mathcal{L}_{CE}(f_S^*(x_{adv}), y), \tag{40}$$

which is maximized with respect to x_{adv}. However, remote sensing images contain structured scene semantics, including roads, runways, buildings, ports, farmland textures, and residential layouts. A perturbation that strongly changes these structures may reduce visual naturalness and weaken cross-model transfer.

To preserve semantic consistency, we introduce a remote sensing manifold consistency loss. Let $\phi(\cdot)$ denote a feature encoder, implemented using the feature extractor of the surrogate model or a remote-sensing pretrained encoder. The manifold consistency loss is

$$\mathcal{L}_{mc} = \|\phi(x_{adv}) - \phi(x)\|_2^2. \tag{41}$$

This term constrains the adversarial example to remain close to the clean image in the semantic embedding space. Unlike a pixel-level constraint alone, $\mathcal{L}_{mc}$ explicitly preserves high-level scene representation. This is important for remote sensing images because semantic identity is often determined by global layout and spatial structure rather than by isolated object appearance.

The adversarial optimization objective is formulated as

$$\max_{x_{adv}} \quad \mathcal{L}_{atk} - \alpha\mathcal{L}_{mc}, \tag{42}$$

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Algorithm 1 Dark-Knowledge-Guided RsMix Surrogate Distillation Attack

Require: Training set $\mathcal{D}$, target model f_T, source model f_S, number of classes C, perturbation budget ϵ, query budget Q, top-k value k, distillation epochs E.

Ensure: Adversarial examples x_{adv}.

Initialize surrogate model $f_S^* \leftarrow f_S$.
Initialize category confusion matrix $C_{mat} \in^{C\times C}$ with zeros.
for $e = 1, \ldots, E$ **do**
  Sample a mini-batch $\{(x_i, y_i)\}$ from $\mathcal{D}$.
  Select category pairs using random sampling in early iterations or $P(i,j) = (C_{mat}/\tau)$ in later iterations.
  Generate RsMix samples $\tilde{x}_{ij}$.
  Query the target model to obtain $p_T(x_i)$ and $p_T(\tilde{x}_{ij})$.
  Compute dark-knowledge gain G_{ij}.
  Update the confusion matrix: $C_{ij} \leftarrow \beta C_{ij} + (1-\beta)G_{ij}$.
  Compute $\mathcal{L}_{KD}^{clean}$, $\mathcal{L}_{KD}^{mix}$, $\mathcal{L}_{cf}$, and $\mathcal{L}_{CE}$.
  Update surrogate parameters by minimizing:

$$\mathcal{L}_{sur} = \lambda_1 \mathcal{L}_{KD}^{clean} + \lambda_2 \mathcal{L}_{KD}^{mix} + \lambda_3 \mathcal{L}_{cf} + \lambda_4 \mathcal{L}_{CE}.$$

end for
for each test image x **do**
  Generate x_{adv} on the distilled surrogate f_S^* by maximizing $\mathcal{L}_{atk} - \alpha \mathcal{L}_{mc}$.
  Project x_{adv} into the L_∞ ball: $\|x_{adv} - x\|_\infty \le \epsilon$.
  Query the target model $f_T(x_{adv})$ to evaluate attack success.
end for
Return adversarial examples and attack results.

subject to

$$\|x_{adv} - x\|_\infty \le \epsilon, \quad x_{adv} \in [0,1]^{H\times W\times 3}, \tag{43}$$

where α controls the strength of manifold preservation.

For iterative attacks, the update rule is

$$x_{adv}^{t+1} = \Pi_\epsilon \left(x_{adv}^t + \eta \cdot \left(\nabla_{x_{adv}^t} [\mathcal{L}_{atk} - \alpha \mathcal{L}_{mc}] \right) \right), \tag{44}$$

where η is the step size and Π_ϵ projects the adversarial image into the L_∞ ball centered at x. For FGSM, a single-step version is used:

$$x_{adv} = \Pi_\epsilon \left(x + \epsilon \cdot (\nabla_x \mathcal{L}_{atk}) \right). \tag{45}$$

The final adversarial examples are evaluated only by querying the target model. No gradient from the target model is used during adversarial generation, satisfying the black-box attack setting. The complete procedure of the proposed method is summarized in Algorithm 1.

IV. Experiments

A. Experimental Setup

1) Datasets: Three publicly available remote sensing scene classification benchmarks of increasing scale are adopted in this work. The AID dataset [20] is a 30-class aerial scene classification benchmark containing 10 000 images at 600×600 pixel resolution, with each category comprising 200 to 400 images captured under varying spatial resolutions and imaging conditions. Sample images are shown in Fig. 4. The NWPU-RESISC45 dataset [21] contains 31 500 images spanning 45 scene classes, with 700 images per class at 256×256 pixels, exhibiting substantial intra-class variation due to diverse geographic regions, viewing angles, and illumination conditions. Sample images are shown in Fig. 5. To assess whether the proposed framework remains effective at a substantially larger data scale, we further employ the MLRSNet dataset [22], which comprises 109 161 optical remote sensing images of 256×256 pixels distributed over 46 scene categories, with each category containing between 1 500 and 3 000 images acquired at multiple spatial resolutions, seasons, and geographic locations. Sample images are shown in Fig. 6. MLRSNet is roughly one order of magnitude larger than AID and more than three times the size of NWPU-RESISC45, and its 46-way label space makes the surrogate distillation problem markedly harder in terms of the number of decision boundaries that must be recovered from a limited query budget. For all three datasets, 80% of the images are used for training and the remaining 20% for evaluation, yielding 8 000/2 000, 25 200/6 300, and 87 328/21 833 training/testing images for AID, NWPU-RESISC45, and MLRSNet, respectively. The split is generated with a fixed random seed so that identical partitions are shared by backbone training, surrogate distillation, and adversarial evaluation. All images are resized to 224×224 pixels through bilinear interpolation to maintain consistency with standard deep learning input requirements.

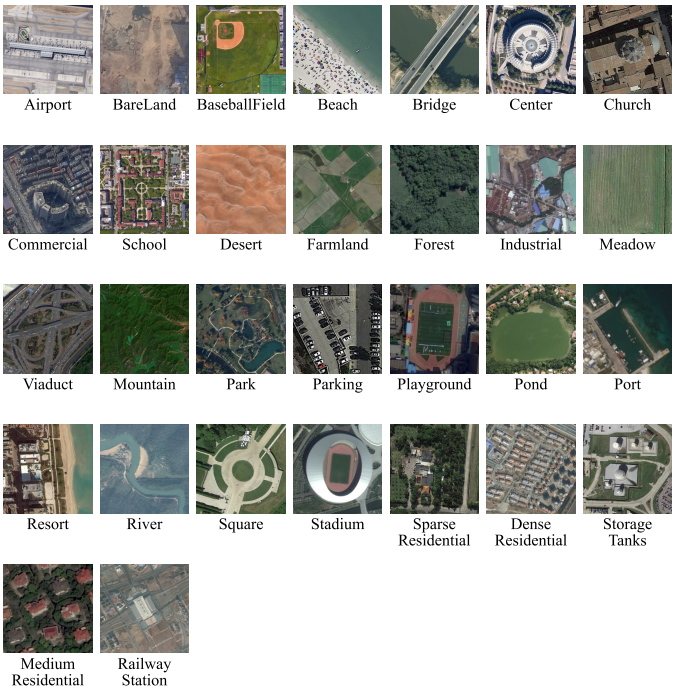

Fig. 4. Example images of each category in the AID dataset, with the category labeled below each image.

2) Model Configurations: To evaluate adversarial transferability, four models are employed as surrogate sources and fourteen as transfer targets, covering convolutional, trans-

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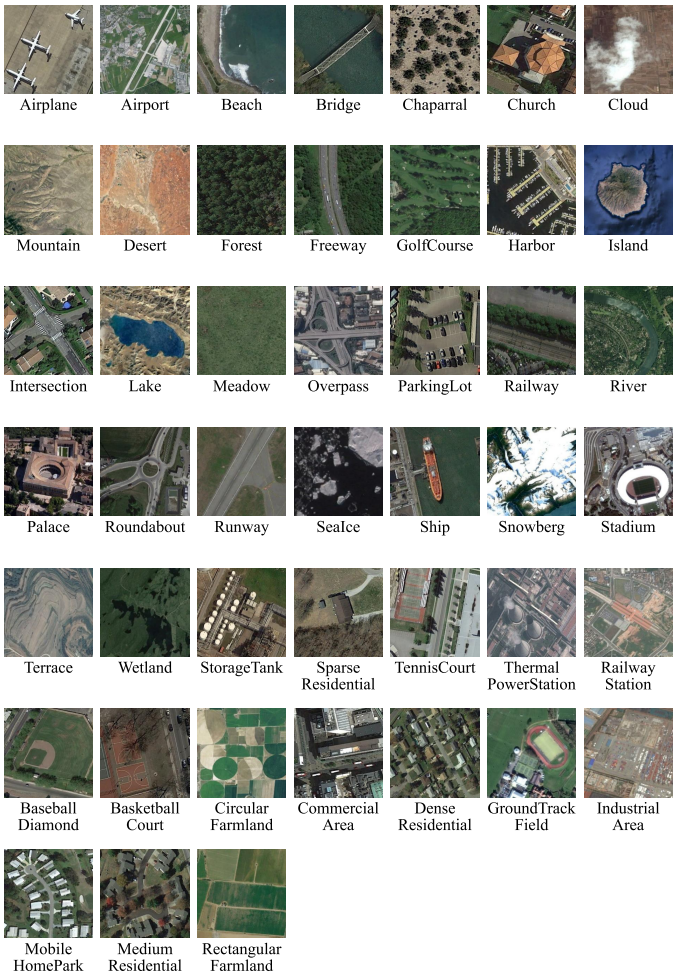

Fig. 5. Example images of each category in the NWPU dataset, with the category labeled below each image.

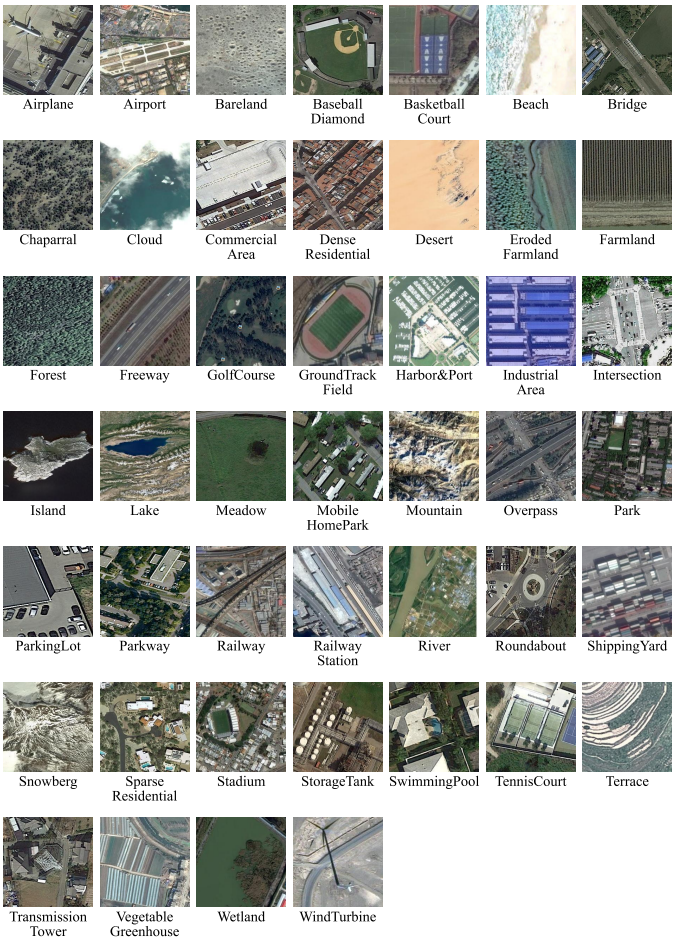

Fig. 6. Example images of each category in the MLRSNet dataset, with the category labeled below each image.

former, hybrid, and state-space architectures. The surrogate models are AlexNet [23], ResNet-18 (Res-18) [24], DenseNet-121 (Dense-121) [25], and RegNetX-400MF (Reg-400) [26], all of which are lightweight networks that can be distilled at low query cost. The target models comprise eight widely adopted backbones, namely VGG-19 [27], ResNet-50 (Res-50) [24], ResNet-101 (Res-101) [24], ResNeXt-50 (ResN-50) [28], RegNetX-8GF (Reg-8GF) [26], RegNetX-16GF (Reg-16GF) [26], ViT-B/16 (ViT-B) [29], and MobileViT-S (MobViT-S) [30], together with six recent architectures, namely ConvNeXt-T [31], Swin-T [32], EfficientNetV2-S (EffV2-S) [33], MaxViT-T [34], MambaOut-T [35], and RSMamba-B [36].

The latter six span the period from 2021 to 2025 and are included so that the evaluation is not restricted to architectures that predate the surrogate design. They represent five families of modern classifiers that would otherwise be unrepresented among the black-box targets: modernized convolutional networks (ConvNeXt-T, EffV2-S), hierarchical window-based transformers (Swin-T), hybrid attention-convolution backbones (MaxViT-T), gated convolutional networks derived from the state-space formulation (MambaOut-T), and a domain-specific state-space backbone designed for remote sensing scene classification (RSMamba-B).

All models are initialized from ImageNet pretrained weights and fine-tuned separately on AID, NWPU-RESISC45, and MLRSNet, so that every architecture-dataset pair corresponds to an independently trained classifier. Fine-tuning uses SGD with momentum 0.9 and weight decay 5×10^{-4}, a batch size of 64, an input resolution of 224×224, and label smoothing of 0.1, with the learning rate scheduled by a one-cycle policy and mixed-precision training enabled. Data augmentation is limited to random horizontal/vertical flipping and random rotation, which preserves the orientation-agnostic nature of overhead imagery without altering scene semantics. All experiments are conducted on a platform equipped with an Intel Core i7-14700F CPU and an NVIDIA GeForce RTX 4080 GPU, running Windows 11 and PyTorch.

The clean test accuracy of all eighteen models is reported in Table I. Every model attains an accuracy between 90.65% and 98.13%, confirming that the targets attacked in the following experiments are well-trained classifiers rather than weak baselines, and that any subsequent misclassification is attributable to the adversarial perturbation rather than to inadequate clean performance. Accuracies on MLRSNet are on average higher than those on AID and NWPU-RESISC45, which is consistent with its larger number of training images per category.

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TABLE I
CLEAN TEST ACCURACY (%) OF THE SURROGATE AND TARGET MODELS ON THE THREE BENCHMARKS

Model	AID	NWPU	MLRSNet
Surrogate models			
AlexNet	93.40	90.65	95.21
Res-18	94.00	93.41	97.27
Dense-121	96.20	94.14	97.89
Reg-400	94.90	93.44	97.39
Target models			
VGG-19	94.85	93.73	96.79
Res-50	95.75	95.68	97.45
Res-101	96.45	95.08	96.02
ResN-50	96.60	96.02	97.71
Reg-8GF	96.75	95.49	97.97
Reg-16GF	96.95	96.37	97.82
ViT-B	96.40	95.84	97.72
MobViT-S	92.70	92.71	96.20
ConvNeXt-T	97.45	96.67	97.41
Swin-T	97.40	97.10	97.59
EffV2-S	97.10	96.87	98.13
MaxViT-T	96.90	96.35	97.05
MambaOut-T	97.15	96.52	97.36
RSMamba-B	97.60	96.94	97.83

3) Comparison Methods: Two families of attacks are used for comparison, since the proposed framework occupies an intermediate position between them: it queries the target model, but only once during surrogate distillation rather than repeatedly during perturbation generation.

The first family consists of nine representative transfer-based attacks, which craft adversarial examples on a white-box source model without querying the target: FGSM [37], I-FGSM [38], SAE [39], ACE [40], AI-FGTM [41], DTA [42], BFA [43], DeCoWA [44], and SFCoT [45]. These methods share the surrogate-to-target transfer setting of the proposed approach and therefore isolate the contribution of the distilled surrogate.

The second family consists of five query-based black-box attacks that interrogate the target model directly during the attack: NES [46], P-RGF [47], SignHunter [48], Square Attack [49], and BASES [50]. NES and SignHunter estimate gradients or gradient signs purely from target-model feedback, Square Attack performs random search over localized square updates, whereas P-RGF and BASES additionally exploit surrogate priors, which makes them the most directly comparable baselines to the proposed method. This family is included so that the query efficiency of the proposed framework is assessed against methods that operate under the same access assumption, rather than only against zero-query transfer attacks.

For all methods, the perturbation budget is fixed at $\epsilon = 25/255$ under the L_∞ norm, so that every attack is evaluated under an identical distortion constraint. For iterative transfer-based methods, the number of iterations is set to 10 with step size $\alpha = \epsilon/10$. For query-based methods, each victim model is attacked on the first 300 correctly classified test images under a budget of 1000 queries per image, and querying terminates as soon as the image is misclassified; the reported query count is therefore the number of queries consumed at the moment of success.

4) Evaluation Metrics: To evaluate the proposed query-efficient black-box attack framework, we use the metrics reported in our experimental tables and figures, including attack success rate, surrogate clean accuracy, target-surrogate agreement, query budget, top-1 probability, and top-k entropy.

Attack success rate is used to measure the effectiveness of the generated adversarial examples. Following the common evaluation protocol for black-box attacks, ASR is computed only on the test samples that are correctly classified by the target model before attack. Let $n_{\text{clean-correct}}$ denote the number of clean test samples correctly classified by the target model, and let $n_{\text{adv-mis}}$ denote the number of these samples that are misclassified after attack. ASR is defined as

$$\text{ASR} = \frac{n_{\text{adv-mis}}}{n_{\text{clean-correct}}} \times 100\%. \tag{46}$$

A higher ASR indicates stronger attack effectiveness.

Surrogate clean accuracy measures the classification performance of the surrogate model on clean test images. It is used to evaluate whether the surrogate still preserves basic remote sensing scene discrimination ability after distillation. It is defined as

$$\text{Acc}_S = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}\left(\arg\max f_S(x_i) = y_i\right) \times 100\%, \tag{47}$$

where f_S denotes the surrogate model, x_i is the i-th clean image, y_i is its ground-truth label, and N is the number of evaluated samples.

Target-surrogate agreement measures the prediction consistency between the surrogate model and the black-box target model on clean test images. It reflects how well the surrogate approximates the target model's decision behavior. The agreement is computed as

$$\text{Agreement} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}\left(\arg\max f_S(x_i) = \arg\max f_T(x_i)\right) \times 100\%, \tag{48}$$

where f_T denotes the target model. A higher agreement indicates that the surrogate better matches the target model's prediction behavior.

Query budget denotes the number of target-model queries allowed for surrogate distillation. In our query-efficiency analysis, different query budgets are used to study how the attack performance changes as more target-model feedback is available. Since each queried sample provides soft-label information from the target model, a smaller query budget with competitive ASR indicates better query efficiency.

Top-1 probability is used in the dark-knowledge visualization to describe the confidence of the target model's most probable prediction. For an input image x, it is defined as

$$P_{\text{top-1}}(x) = \max_c p_T(c|x), \tag{49}$$

where $p_T(c|x)$ is the softmax probability of class c predicted by the target model. A very high top-1 probability usually indicates a peaked distribution with limited dark knowledge.

Top-k entropy is used to quantify the amount of dark knowledge contained in the target model's response. Given an input image x, let $\Omega_k(x)$ denote the set of top-k predicted

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TABLE II
ASR (%) OF TEN BLACK-BOX ATTACK METHODS ON THE AID DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN GREEN, AND THE WORST RESULTS ARE HIGHLIGHTED IN RED.

Surrogate Models	Methods	Target Models							
		CNNs						Transformers	
		VGG-19	Res-50	Res-101	ResN-50	Reg-8GF	Reg-16GF	ViT-B	MobViT-s
Alexnet	FGSM	55.30	60.50	74.20	44.40	42.90	51.33	30.90	47.00
	I-FGSM	57.97	61.30	74.73	45.50	44.73	52.20	31.70	49.67
	SAE	60.03	65.87	79.97	47.87	47.90	54.60	35.40	52.63
	ACE	58.50	64.60	79.10	47.00	45.90	53.70	34.40	50.20
	AI-FGTM	61.30	66.20	80.73	49.00	49.10	53.90	37.00	57.10
	DTA	59.70	66.03	80.00	48.13	48.20	54.10	35.23	53.43
	BFA	59.97	65.70	80.03	48.03	47.63	54.40	35.07	52.27
	DeCoWA	60.07	65.97	79.93	48.37	48.67	54.97	35.77	53.07
	SFCoT	56.40	60.60	74.90	44.90	43.70	51.70	32.00	48.60
	Ours	67.63	46.94	46.72	46.03	45.99	44.85	44.87	71.52
Resnet-18	FGSM	55.20	60.43	74.23	44.40	42.90	51.30	30.80	46.93
	I-FGSM	58.67	60.10	76.00	45.50	46.50	52.40	31.73	50.40
	SAE	59.93	65.60	79.87	47.93	47.80	54.30	35.53	52.10
	ACE	58.10	63.70	78.03	46.20	45.60	52.57	34.30	49.80
	AI-FGTM	61.97	67.13	81.97	49.60	51.70	55.07	36.00	56.77
	DTA	60.03	66.20	80.10	48.27	48.03	54.23	35.10	52.57
	BFA	60.30	66.83	80.40	50.03	49.80	54.63	35.03	53.80
	DeCoWA	60.50	67.13	80.63	49.23	50.13	55.40	35.87	53.83
	SFCoT	56.50	61.40	74.90	45.20	43.70	52.20	31.00	47.80
	Ours	78.70	77.21	76.93	75.89	58.27	64.24	30.45	78.69
Densenet121	FGSM	55.20	60.43	74.20	44.47	42.90	51.30	30.70	46.90
	I-FGSM	57.50	60.40	75.47	44.83	44.83	52.77	31.50	49.10
	SAE	59.87	65.73	79.63	47.77	48.00	54.10	35.37	52.37
	ACE	58.50	63.10	78.50	46.13	46.30	53.00	34.00	50.70
	AI-FGTM	62.17	66.80	81.07	49.17	50.70	54.83	35.37	55.53
	DTA	59.87	66.43	80.17	48.77	48.83	54.67	34.97	52.40
	BFA	60.60	66.80	80.77	49.47	50.20	55.47	35.20	53.93
	DeCoWA	60.73	66.93	81.00	49.53	50.20	55.83	35.60	53.90
	SFCoT	56.50	60.70	74.90	45.10	44.50	52.00	30.90	47.40
	Ours	75.70	82.54	71.72	81.59	70.60	73.80	33.87	81.61
Regnet_x_400mf	FGSM	55.30	60.40	74.20	44.40	42.90	51.30	30.70	47.00
	I-FGSM	57.07	60.20	75.33	44.50	44.60	51.13	31.10	48.43
	SAE	60.37	65.80	79.73	47.87	47.73	54.67	35.33	51.93
	ACE	58.23	64.10	78.00	46.03	46.40	52.57	34.20	49.43
	AI-FGTM	60.20	65.33	80.40	47.50	49.73	53.90	35.23	54.77
	DTA	59.43	65.33	79.60	47.33	47.83	53.57	34.83	51.57
	BFA	59.30	65.70	79.50	47.60	48.50	53.80	34.60	51.70
	DeCoWA	59.77	66.40	80.03	48.23	48.93	54.23	35.83	53.37
	SFCoT	56.02	61.18	75.43	45.59	45.19	52.30	31.30	48.60
	Ours	58.78	48.25	48.70	44.51	66.65	54.37	32.21	64.24

classes of the target model. The probabilities within the top-k set are normalized as

$$\hat{p}_T(c|x) = \frac{p_T(c|x)}{\sum_{j\in\Omega_k(x)} p_T(j|x)}, \quad c \in \Omega_k(x). \tag{50}$$

The top-k entropy is then defined as

$$H_k(x) = -\sum_{c\in\Omega_k(x)} \hat{p}_T(c|x)\log \hat{p}_T(c|x). \tag{51}$$

A larger top-k entropy means that the target model assigns meaningful probabilities to more classes, indicating richer dark knowledge and stronger inter-class relation information.

B. Experimental Results and Analysis

1) Quantitative Results: Tables II–IV report the ASR results over 32 surrogate–target combinations on each dataset. The proposed method achieves average ASRs of 60.75%, 63.21%, and 61.19% on AID, NWPU-RESISC45, and MLRSNet, and ranks first in 17, 26, and 18 of the 32 settings. On AID and NWPU-RESISC45 it outperforms the strongest baseline by 3.96 and 12.20 percentage points, whereas on

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TABLE III
ASR (%) OF TEN BLACK-BOX ATTACK METHODS ON THE NWPU-RESISC45 DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN GREEN, AND THE WORST RESULTS ARE HIGHLIGHTED IN RED.

Surrogate Models	Methods	Target Models							
		CNNs						Transformers	
		VGG-19	Res-50	Res-101	ResN-50	Reg-8GF	Reg-16GF	ViT-B	MobViT-s
Alexnet	FGSM	40.57	43.77	59.07	40.53	31.17	44.77	56.00	43.40
	I-FGSM	41.57	46.00	59.67	41.43	32.53	45.57	56.37	45.17
	SAE	43.27	47.30	62.67	44.53	33.40	49.80	60.00	47.30
	ACE	44.60	48.00	64.30	44.80	33.30	50.30	61.50	47.70
	AI-FGTM	45.30	50.70	65.00	46.80	35.80	50.90	62.60	48.90
	DTA	44.33	48.83	63.57	46.10	34.03	50.70	61.80	47.90
	BFA	44.80	49.33	64.40	46.23	34.00	50.47	61.37	47.80
	DeCoWA	44.97	49.63	64.23	46.67	34.27	51.00	62.23	48.43
	SFCoT	41.10	45.70	59.90	42.70	32.60	46.10	56.40	44.30
	Ours	77.76	68.10	56.98	59.07	53.91	56.97	53.33	75.47
Resnet-18	FGSM	40.53	43.70	59.00	40.50	31.13	44.70	56.00	43.37
	I-FGSM	41.33	45.37	60.27	42.23	31.97	45.73	56.27	44.67
	SAE	43.07	47.43	62.43	44.27	33.87	50.30	59.60	47.57
	ACE	44.17	48.80	62.90	44.90	33.30	50.20	61.90	47.70
	AI-FGTM	44.67	52.30	65.83	48.13	36.07	51.20	61.50	49.23
	DTA	44.23	49.87	65.10	46.63	34.53	50.57	61.47	48.00
	BFA	44.77	51.47	65.33	47.47	35.17	50.77	61.30	48.63
	DeCoWA	45.27	51.50	65.83	47.20	35.57	51.57	61.63	49.30
	SFCoT	41.20	45.50	59.90	42.70	32.20	45.09	56.10	44.30
	Ours	70.45	84.75	76.60	73.12	56.27	61.81	29.08	74.30
Densenet121	FGSM	40.53	43.73	59.00	40.60	31.10	44.70	56.00	43.27
	I-FGSM	40.93	45.17	59.27	42.13	31.53	45.47	55.93	44.30
	SAE	42.90	47.13	62.47	44.23	33.60	49.93	59.70	47.33
	ACE	43.50	48.60	63.10	44.83	32.50	50.10	61.10	48.30
	AI-FGTM	45.60	52.77	66.00	49.00	36.07	52.50	61.47	50.57
	DTA	45.00	50.30	64.83	47.17	33.87	51.10	61.00	48.23
	BFA	45.03	51.60	65.77	48.13	35.53	51.67	61.37	49.57
	DeCoWA	45.63	51.80	66.00	48.20	35.87	51.83	61.47	50.17
	SFCoT	40.80	45.10	59.10	42.50	32.92	45.40	56.00	44.50
	Ours	71.09	85.07	76.84	81.54	66.91	74.34	35.17	79.13
Regnet_x_400mf	FGSM	40.50	43.60	59.03	40.50	31.10	44.73	56.00	43.27
	I-FGSM	40.87	44.33	59.30	41.07	31.43	45.23	56.03	43.47
	SAE	42.77	47.03	62.43	44.30	33.70	50.10	60.00	48.07
	ACE	43.90	47.93	62.80	44.60	33.70	49.80	60.60	47.33
	AI-FGTM	44.57	51.10	64.50	46.50	35.20	51.00	61.50	49.03
	DTA	44.37	48.80	63.97	45.67	33.90	50.27	61.03	47.83
	BFA	44.10	48.80	63.90	45.40	33.50	50.17	61.20	47.80
	DeCoWA	44.73	50.47	65.07	46.67	35.23	51.23	61.83	48.67
	SFCoT	40.90	44.80	59.30	42.13	32.30	45.80	56.00	43.89
	Ours	55.36	55.23	50.70	48.43	62.73	55.47	31.49	65.37

MLRSNet its overall mean falls 3.58 points below DeCoWA (64.77%). We report this deficit explicitly because its structure is informative: it is produced entirely by two of the four surrogates.

Separating the MLRSNet results by surrogate makes this clear. With ResNet-18 and DenseNet-121 the proposed method attains 71.55% and 70.86%, wins 12 of the corresponding 16 cells, and is effectively level with DeCoWA when pooled (71.21% versus 71.42%). With AlexNet and RegNetX-400MF it reaches only 54.41% and 47.95%, trailing DeCoWA by 7.22 and 6.64 points. The same ordering holds on the smaller benchmarks, where DenseNet-121 attains the highest averages of 71.43% and 71.26%. Averaged over the three datasets, the two expressive surrogates exceed the two lightweight ones by roughly twenty percentage points, while the nine baselines vary by under three points across the same four sources. The framework is therefore markedly more surrogate-dependent than the baselines, as expected for a method that actively reshapes the surrogate towards the victim; Section V analyses the mechanism.

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TABLE IV
ASR (%) OF TEN BLACK-BOX ATTACK METHODS ON THE MLRSNET DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN GREEN, AND THE WORST RESULTS ARE HIGHLIGHTED IN RED.

Surrogate Models	Methods	Target Models							
		CNNs						Transformers	
		VGG-19	Res-50	Res-101	ResN-50	Reg-8GF	Reg-16GF	ViT-B	MobViT-s
AlexNet	FGSM	48.25	43.21	48.18	43.34	35.14	42.27	35.91	52.59
	I-FGSM	51.98	46.31	49.45	46.54	40.05	44.42	39.48	55.30
	SAE	57.33	51.52	53.22	51.06	45.34	49.52	45.19	59.89
	ACE	59.85	54.51	56.92	54.12	48.09	51.51	47.95	62.45
	AI-FGTM	66.46	62.65	63.12	60.83	55.68	56.83	51.60	68.69
	DTA	70.65	61.44	62.16	59.05	54.14	58.54	50.10	71.14
	BFA	71.47	61.13	62.29	59.36	53.79	58.30	48.87	70.23
	DeCoWA	72.13	61.13	62.21	59.07	54.59	60.32	51.25	72.36
	SFCoT	62.05	54.55	57.11	53.97	48.85	49.70	46.19	63.73
	Ours	79.18	38.00	38.65	53.33	57.59	36.77	54.87	76.93
ResNet-18	FGSM	50.15	45.87	40.97	44.44	35.43	39.32	48.68	50.51
	I-FGSM	52.97	45.02	43.18	47.72	39.30	42.59	46.99	53.93
	SAE	58.03	50.69	49.67	53.65	45.14	48.03	49.76	58.39
	ACE	60.98	53.60	52.44	56.75	49.41	51.65	51.46	60.36
	AI-FGTM	69.68	67.03	72.19	64.40	58.22	61.98	53.96	67.96
	DTA	72.70	70.28	66.27	67.25	62.16	65.18	52.68	71.39
	BFA	75.74	73.33	69.01	70.41	65.10	68.56	52.71	73.48
	DeCoWA	77.55	75.21	71.42	71.80	66.87	70.96	52.68	74.38
	SFCoT	63.31	60.33	56.10	59.58	51.85	54.19	49.59	62.35
	Ours	84.50	80.30	68.70	79.10	74.40	77.90	25.30	82.20
DenseNet-121	FGSM	49.54	50.77	45.07	49.48	43.49	46.37	35.43	51.86
	I-FGSM	52.66	50.63	49.13	53.39	47.79	50.37	38.86	55.68
	SAE	57.08	58.75	53.51	57.63	52.04	54.94	41.43	59.75
	ACE	60.19	61.90	56.32	60.99	55.82	58.02	43.56	62.52
	AI-FGTM	69.29	70.94	65.73	69.05	63.52	67.44	45.37	69.39
	DTA	72.55	74.59	69.81	72.95	67.89	71.40	46.93	73.54
	BFA	74.85	77.34	73.32	76.00	71.11	74.22	45.39	76.06
	DeCoWA	76.72	78.54	75.26	77.84	73.42	76.47	45.70	77.88
	SFCoT	63.13	64.18	59.14	63.25	57.41	59.73	40.97	65.13
	Ours	82.47	81.34	40.98	80.94	76.70	79.44	44.88	80.16
RegNet_X_400MF	FGSM	42.26	32.28	32.15	35.63	48.43	47.25	29.24	34.24
	I-FGSM	45.92	36.18	34.30	39.71	51.52	49.14	30.14	38.62
	SAE	53.05	43.22	41.69	46.57	57.08	55.78	36.35	45.85
	ACE	50.31	41.02	39.20	44.61	55.29	53.11	34.60	43.79
	AI-FGTM	52.81	48.66	47.62	51.21	61.40	55.76	35.29	50.99
	DTA	51.28	51.22	48.60	54.18	64.25	54.97	37.52	53.35
	BFA	51.40	53.13	50.59	56.25	66.18	55.70	38.53	55.53
	DeCoWA	52.97	54.85	52.06	58.36	67.01	55.25	39.05	57.18
	SFCoT	48.12	39.04	37.14	42.81	53.34	51.87	33.09	41.85
	Ours	52.68	31.04	31.03	52.15	69.39	55.65	31.50	60.14

The target side is more stable. ViT-B remains the hardest target across all three datasets, with average ASRs of 35.35%, 37.27%, and 39.14%, whereas MobileViT-S reaches 74.02%, 73.57%, and 74.86%, suggesting that global self-attention weakens transfer from convolutional surrogates while hybrid structures retain transferable local features. On MLRSNet the method still leads on VGG-19, RegNetX-8GF, and MobileViT-S by 4.86, 4.05, and 4.41 points. One column deviates: on ResNet-101 it averages 44.84% against 65.24% for the best baseline, and DenseNet-121 reaches only 40.98% there despite 81.34% on ResNet-50. This column, together with ViT-B, accounts for the gap between DenseNet-121's six column-wise wins and its below-baseline mean; excluding both, the pooled ASR of ResNet-18 and DenseNet-121 on MLRSNet is 79.95% against 74.80%. We report the ResNet-101 outlier as an open observation rather than offering a post hoc explanation. Overall, the framework scales to a benchmark one order of magnitude larger with a 46-way label space, but its advantage is conditional on the expressiveness of the surrogate.

2) Evaluation on Recent Architectures: Table V reports the ASR on six targets spanning 2021 to 2025: ConvNeXt-T, Swin-T, EfficientNetV2-S, MaxViT-T, MambaOut-T, and

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TABLE V
ASR (%) OF TEN BLACK-BOX ATTACK METHODS AGAINST SIX MODERN ARCHITECTURES ON THE AID DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD.

Method	AlexNet						ResNet-18					
	CNXt-T	Swin-T	EffV2-S	MaxViT-T	MbOut-T	RSMb-B	CNXt-T	Swin-T	EffV2-S	MaxViT-T	MbOut-T	RSMb-B
FGSM	49.77	26.27	51.43	39.95	48.32	28.94	49.68	26.18	51.37	39.89	48.26	28.85
I-FGSM	52.17	26.95	52.11	42.22	50.64	29.71	52.80	26.97	51.09	42.84	50.92	29.63
SAE	54.03	30.09	55.99	44.74	52.71	32.15	53.94	30.20	55.76	44.29	52.58	32.24
ACE	52.65	29.24	54.91	42.67	51.09	31.06	52.29	29.16	54.15	42.33	50.83	30.97
AI-FGTM	55.17	31.45	**56.27**	48.54	**53.86**	**33.28**	55.77	**30.60**	57.06	48.25	54.14	33.05
DTA	53.73	29.95	56.13	45.42	52.44	32.07	54.03	29.84	56.27	44.68	52.76	32.18
BFA	53.97	29.81	55.85	44.43	52.68	31.85	54.27	29.78	56.81	45.73	53.09	31.94
DeCoWA	54.06	30.40	56.07	45.11	52.95	32.44	54.45	30.49	57.06	45.76	53.31	32.56
SFCoT	50.76	27.20	51.51	41.31	49.15	29.38	50.85	26.35	52.19	40.63	49.47	29.26
Ours	**60.87**	**38.14**	39.90	**60.79**	47.62	31.86	**70.83**	25.88	**65.63**	**66.89**	**62.47**	**39.85**
Method	DenseNet-121						RegNetX-400MF					
	CNXt-T	Swin-T	EffV2-S	MaxViT-T	MbOut-T	RSMb-B	CNXt-T	Swin-T	EffV2-S	MaxViT-T	MbOut-T	RSMb-B
FGSM	49.68	26.10	51.37	39.87	48.21	28.82	49.77	26.10	51.34	39.95	48.35	28.79
I-FGSM	51.75	26.78	51.34	41.74	50.38	29.55	51.36	26.44	51.17	41.17	50.06	29.42
SAE	53.88	30.06	55.87	44.51	52.63	32.11	**54.33**	30.03	55.93	44.14	52.87	32.19
ACE	52.65	28.90	53.64	43.10	51.24	30.85	52.41	29.07	54.49	42.02	51.16	30.94
AI-FGTM	55.95	30.06	56.78	47.20	54.32	33.14	54.18	29.95	55.53	46.55	**53.11**	32.68
DTA	53.88	29.72	56.47	44.54	52.71	32.02	53.49	29.61	55.53	43.83	52.34	31.97
BFA	54.54	29.92	56.78	45.84	53.16	31.90	53.37	29.41	55.85	43.95	52.19	31.73
DeCoWA	54.66	**30.26**	56.89	45.82	53.42	32.51	53.79	**30.46**	**56.44**	45.36	52.68	**32.87**
SFCoT	50.85	26.27	51.60	40.29	49.32	29.19	50.42	26.61	52.00	41.31	49.58	29.44
Ours	**68.13**	28.79	**70.16**	**69.37**	**63.92**	**41.23**	52.90	27.38	41.01	**54.60**	46.18	29.94

RSMamba-B. Over the 24 surrogate–target combinations the proposed method ranks first in 14 and averages 50.18%, exceeding by 4.06 percentage points the 46.12% of AI-FGTM, the strongest baseline overall here. As in Tables II–IV, each per-target comparison below is made against the strongest baseline in that column.

The advantage is concentrated on targets retaining a convolutional stem. On ConvNeXt-T, MaxViT-T, and MambaOut-T the method attains 63.18%, 62.91%, and 55.05%, leading AI-FGTM by 7.91, 15.28, and 1.19 points. Notably, MambaOut-T derives from the state-space formulation but removes the selective-scan operator, so its gradients remain compatible with those of a convolutional surrogate. Conversely, the two targets that aggregate evidence globally give the lowest values in the table: 35.72% on RSMamba-B, still 2.68 points above AI-FGTM, and 30.05% on Swin-T, which falls 0.47 points *below* it. The Swin-T shortfall reflects surrogate variance rather than a uniform weakness: the method wins that cell under AlexNet (38.14% against 31.45%) but drops to 25.88%, 28.79%, and 27.38% under the other three surrogates. Shifted-window attention and bidirectional state-space scanning both average out much of the localized perturbation energy produced from a convolutional surrogate; that two unrelated lineages share this deficit points to the aggregation mechanism rather than the transformer formulation as the limiting factor.

EfficientNetV2-S illustrates the surrogate dependence dominating the table: 65.63% and 70.16% with ResNet-18 and DenseNet-121 against 39.90% and 41.01% with AlexNet and RegNetX-400MF, leaving its average of 54.17% 2.44 points below DeCoWA (56.61%). Pooled over the six targets, the two expressive surrogates reach 55.26% and 56.93% and win 10 of their 12 cells, the two lightweight ones 46.53% and 42.00% and win 4 of 12. The method thus leads the strongest baseline on four of the six targets and remains effective against classifiers introduced up to 2025, including RSMamba-B designed for remote sensing, trailing only on Swin-T and EfficientNetV2-S. Both deficits reflect the two conditions analysed in Section V: global evidence aggregation in the target, and limited surrogate expressiveness.

TABLE VI
VISUAL QUALITY COMPARISON ON THE AID DATASET, AVERAGED OVER THE FOUR SURROGATE MODELS. ALL METHODS USE THE SAME PERTURBATION BUDGET $\epsilon_\infty = 25/255$ AND ARE EVALUATED ON THE SAME 1008 TEST IMAGES. HIGHER PSNR AND SSIM, AND LOWER LPIPS, INDICATE BETTER VISUAL SIMILARITY TO THE CLEAN IMAGES.

Method	PSNR↑	SSIM↑	LPIPS↓
FGSM	24.07	0.5504	0.5039
I-FGSM	33.17	0.8853	0.1670
SAE	28.92	0.9738	0.1270
ACE	25.19	0.9470	0.1336
AI-FGTM	28.51	0.7482	0.3080
DTA	26.24	0.6573	0.3790
BFA	26.36	0.6690	0.4051
DeCoWA	26.54	0.6881	0.3444
SFCoT	26.74	0.6744	0.3705
Ours	27.01	0.6752	0.3340

3) Visual Quality Analysis: Table VI compares the visual quality of the adversarial examples under the same ℓ_∞ perturbation budget. I-FGSM achieves the highest PSNR of 33.17 dB, while SAE obtains the highest SSIM and the lowest LPIPS, indicating the best overall perceptual similarity among the compared methods. The proposed method achieves a PSNR of 27.01 dB, an SSIM of 0.6752, and an LPIPS of 0.3340. Compared with FGSM, it improves PSNR by 2.94 dB,

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TABLE VII
COMPUTATIONAL EFFICIENCY ON AID, AVERAGED OVER FOUR SURROGATE ARCHITECTURES. TOTAL RUNTIME IS MEASURED ON 1,008 IMAGES AND INCLUDES THE ONE-TIME DISTILLATION STAGE FOR OUR METHOD. ADVERSARIAL-EXAMPLE GENERATION TIME EXCLUDES MODEL LOADING, DATA LOADING, WARM-UP, AND DISTILLATION. RESOURCE CONSUMPTION IS PEAK ALLOCATED GPU MEMORY.

Method	Runtime (s)↓	Generation (ms/image)↓	Memory (MB)↓
FGSM	2.24	2.22	838
I-FGSM	15.67	15.54	856
SAE	14.07	13.96	252
ACE	279.65	277.43	1440
AI-FGTM	16.20	16.07	884
DTA	152.82	151.61	931
BFA	77.40	76.79	1407
DeCoWA	451.52	447.94	881
SFCoT	47.10	46.73	876
Ours	183.18	65.06	2022

TABLE VIII
COMPARISON WITH QUERY-BASED BLACK-BOX ATTACKS ON AID. ALL METHODS USE $\epsilon_\infty = 25/255$. EACH VICTIM IS EVALUATED ON THE FIRST 300 CORRECTLY CLASSIFIED TEST IMAGES. THE PER-IMAGE QUERY BUDGET IS 1000, WHILE BASES USES ITS OFFICIAL CAP OF 50.

		VGG-19		Res-50		Reg-16GF		ViT-B/16	
Method	Venue	ASR	#q	ASR	#q	ASR	#q	ASR	#q
NES	ICML'18	39.4	522	35.1	531	31.7	513	25.9	575
P-RGF	NeurIPS'19	65.8	384	69.4	343	54.6	401	29.8	473
SignHunter	ICLR'20	88.6	93	86.9	78	84.1	95	72.5	153
Square	ECCV'20	92.7	38	91.3	71	89.5	69	78.6	98
BASES	NeurIPS'22	84.9	2	82.7	1	75.6	2	64.2	4
Ours	–	78.7	64†	82.5	64†	73.8	64†	44.9	64†

† The proposed method spends 19,200 queries once per victim during distillation, corresponding to 64 amortized queries per image over 300 images. Over the full test set of 2,000 images, this becomes 9.6 queries per image.

increases SSIM by 0.1248, and reduces LPIPS by 0.1699.

Although the proposed method does not achieve the best value for every visual-quality metric, its results remain comparable to those of several transfer-based attacks under the identical perturbation constraint. This reflects the trade-off between visual fidelity and attack transferability: the proposed method is designed primarily to improve black-box ASR through surrogate distillation and dark-knowledge guidance, rather than to optimize perceptual similarity alone. The results therefore show that the proposed attack maintains reasonable visual similarity while pursuing stronger cross-model transferability.

4) Computational Efficiency Analysis: As shown in Table VII, the proposed method requires 65.06 ms to generate an adversarial example, which is lower than the generation time of ACE, DTA, BFA, and DeCoWA. However, it incurs a one-time distillation cost of approximately 117.60 s and has the highest peak memory consumption of 2,022 MB. Thus, the proposed method is more suitable for repeated attacks that require continuous generation of a large number of adversarial examples, since the initial distillation cost can be amortized over multiple images. For small-scale, one-time attacks, low-cost baselines remain more appropriate.

Regarding deployment under resource constraints, the peak requirement of 2,022 MB is modest for a single consumer-grade GPU and the method runs without modification on the RTX 4080 used in our experiments; memory is therefore not the limiting factor on desktop-class hardware. The binding constraint is instead the distillation stage, which requires backpropagation through the surrogate and holds both the surrogate and the query batch in memory, and which must be repeated whenever the victim model changes. On embedded or edge platforms with limited memory and no training capability, this stage rather than the 65.06 ms per-image generation is the bottleneck, and the lightweight baselines in Table VII such as FGSM and I-FGSM remain the more practical choice. A natural mitigation is to perform distillation once on a resource-sufficient machine and to deploy only the distilled surrogate for inference-time perturbation generation, which reduces the on-device cost to a single forward-backward pass through a lightweight network.

5) Comparison with Query-Based Black-Box Attacks: Table VIII compares the proposed method with representative query-based black-box attacks under the same perturbation constraint. The proposed method achieves an average ASR of 69.98% across the four victim models, substantially outperforming NES and P-RGF by 36.95 and 15.08 percentage points, respectively. Its amortized query cost is only 64 queries per image over the evaluated 300 images, compared with 535.25 and 400.25 queries per image for NES and P-RGF.

SignHunter, Square, and BASES obtain higher average ASRs of 83.03%, 88.03%, and 76.85%, respectively, because they adaptively query the victim for each image. However, their query costs are different from that of the proposed method. *Ours* requires fewer queries than SignHunter and Square, while BASES achieves the lowest query cost with a slight ASR advantage. The proposed method spends 19,200 queries only once for each victim; this cost decreases to 9.6 queries per image when amortized over 2,000 test images. Therefore, *Ours* provides a complementary trade-off: it sacrifices some per-image ASR compared with strong adaptive query attacks in exchange for a nearly constant query cost when a large number of adversarial examples must be generated.

6) Grad-CAM Visualization Analysis: Fig. 7 shows the Grad-CAM visualization before and after attack. For the clean image, the target model focuses on the discriminative land-cover structures corresponding to the correct category. After adding the adversarial perturbation, the image remains visually similar, but the activation region shifts to irrelevant or misleading areas, resulting in misclassification. This indicates that the proposed method not only changes the final prediction but also alters the spatial evidence used by the target model. The result confirms that the surrogate model distilled with RsMix and top-k dark knowledge can generate transferable perturbations that effectively manipulate the decision behavior of black-box remote sensing classifiers.

7) Ablation Study of the Proposed Components: This subsection investigates the contribution of each component in the proposed dark-knowledge surrogate distillation framework. Starting from a non-distilled source model, we progressively introduce top-k dark-knowledge distillation, RsMix mixed-sample distillation, confusion-matrix-guided pairing,

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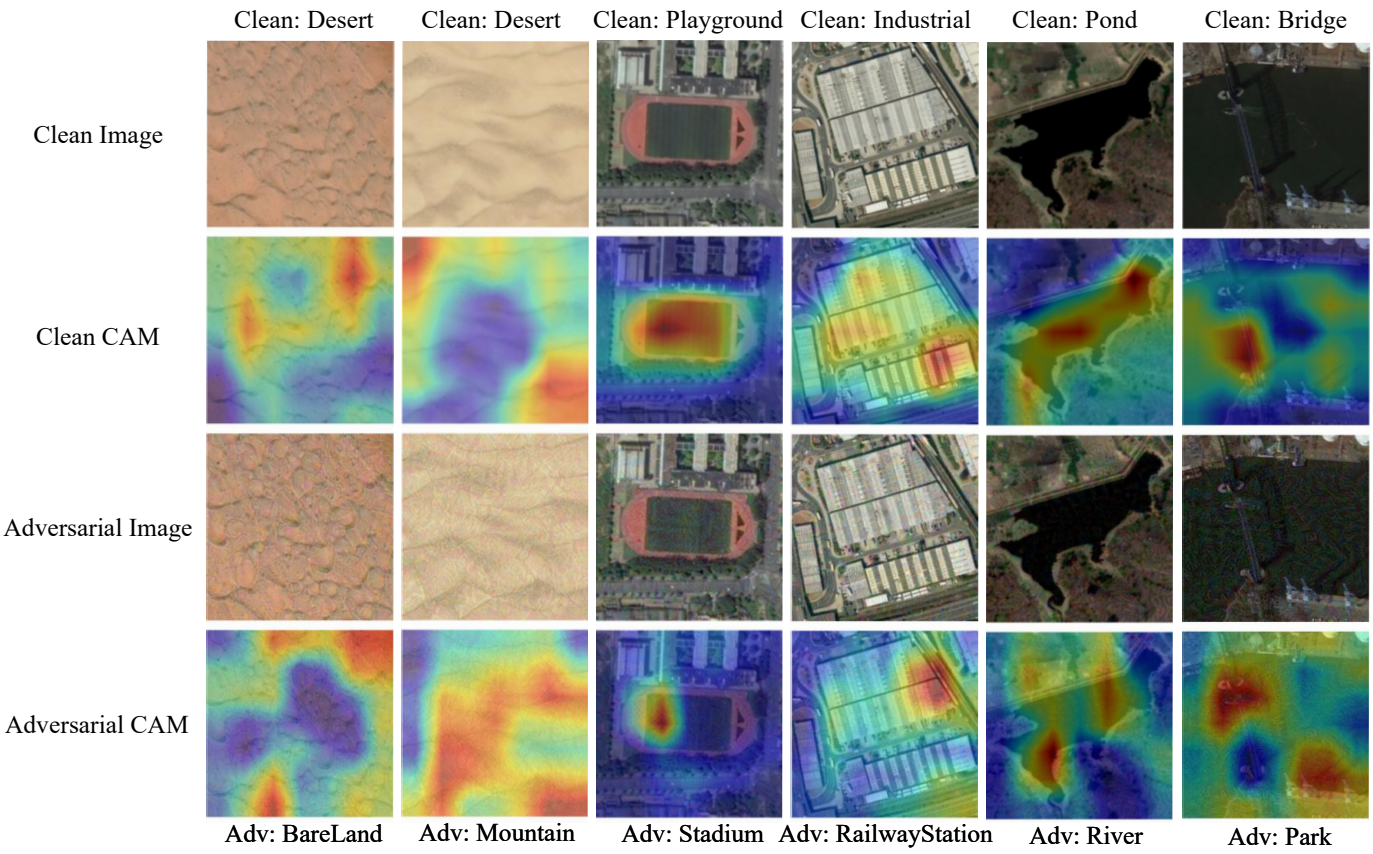

Fig. 7. Grad-CAM visualization before and after attack. The clean image is correctly classified by the target model, and the activation map focuses on the discriminative land-cover structures. After attack, the image remains visually similar, but the activation region shifts to misleading areas, causing the target model to output an incorrect label.

TABLE IX
ABLATION STUDY OF EACH COMPONENT IN THE PROPOSED DARK-KNOWLEDGE SURROGATE DISTILLATION FRAMEWORK

Variant	RsMix	Confusion	$\mathcal{L}_{cf}$	Surrogate Acc (%)	Agreement (%)	ASR (%)
Source only	–	–	–	93.67	93.75	68.94
+ Top-k KD	–	–	–	93.28	93.28	70.75
+ Random RsMix	yes	–	–	92.81	92.81	72.23
+ RsMix	yes	yes	–	92.73	92.73	72.64
+ $\mathcal{L}_{cf}$	yes	yes	yes	92.50	92.66	72.88
Full	yes	yes	yes	92.50	92.66	72.88

and the classification feature alignment loss $\mathcal{L}_{cf}$. To make the component-level contributions observable, all variants share the same MI-FGSM attack backbone, so that performance differences originate only from the quality of the distilled surrogate rather than from a stronger attack generator.The experimental results are summarized in Table IX.

As shown in Table IX, the ASR increases consistently as more components are incorporated, rising from 68.94% for the non-distilled source model to 72.88% for the full configuration. The first gain comes from top-k dark-knowledge distillation, which improves ASR by 1.81 percentage points over the source-only baseline. This confirms that querying the black-box target and transferring its top-k soft responses to the surrogate produces a more target-consistent decision boundary than relying on the original source model. Introducing RsMix mixed-sample distillation yields a further improvement to 72.23%, indicating that mixed remote sensing samples expose additional inter-class relations that are not visible from clean images alone. Replacing random pairing with confusion-matrix-guided pairing brings an additional gain to 72.64%, which demonstrates that prioritizing semantically confusable category pairs increases the dark-knowledge density of each query and benefits surrogate learning. Finally, the classification feature alignment loss $\mathcal{L}_{cf}$ pushes the ASR to 72.88%, verifying that aligning the surrogate feature representation with the target-induced class prototype provides decision-boundary-relevant information rather than acting as a generic regularizer.

It is worth noting that the surrogate clean accuracy decreases slightly from 93.67% to 92.50% as more distillation components are added. This is expected, because the objective of

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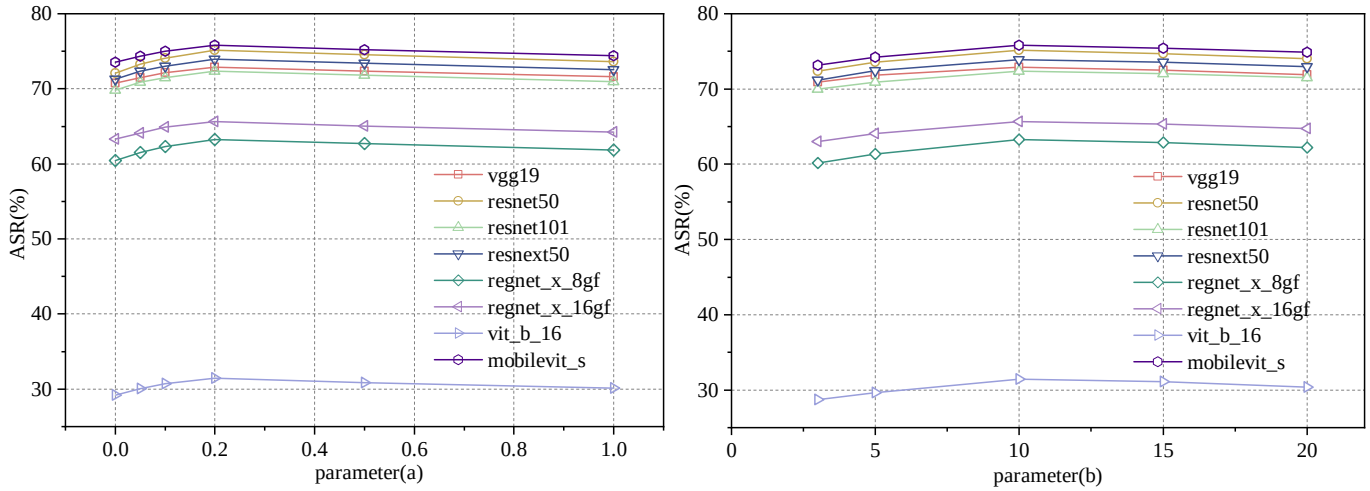

Fig. 8. Parameter sensitivity analysis on different target models. (a) Sensitivity to the classification feature alignment weight λ_{cf}. (b) Sensitivity to the top-k value used in dark-knowledge distillation. The results show that $\lambda_{cf} = 0.2$ and $k = 10$ achieve the best overall performance across different architectures.

the surrogate is not to maximize classification accuracy but to better approximate the target model's decision behavior for transferable attack generation. The fact that ASR improves while clean accuracy drops indicates that the proposed components reshape the surrogate decision boundary toward the target model rather than toward the ground-truth labels, which is precisely the desired behavior in a query-efficient black-box attack scenario.

8) Parameter Sensitivity Analysis: We analyze the sensitivity of two key hyperparameters: the weight of the classification feature alignment loss λ_{cf} and the number of selected dark-knowledge classes k. As shown in Fig. 8, both parameters exhibit stable trends across different target models.

For λ_{cf}, the ASR is relatively lower when $\lambda_{cf} = 0$, indicating that output-level distillation alone is insufficient to fully capture the target model's class-relation structure. As λ_{cf} increases from 0 to 0.2, the ASR consistently improves, showing that moderate feature alignment helps the surrogate model learn target-induced class prototypes and improves adversarial transferability. However, when λ_{cf} is further increased to 0.5 or 1.0, the ASR slightly decreases, suggesting that overly strong feature constraints may reduce the flexibility of the surrogate feature space. Therefore, we set $\lambda_{cf} = 0.2$ in the main experiments.

For the top-k value, a small k provides only limited non-top-1 class information, which weakens the surrogate model's ability to learn the target model's inter-class relations. Increasing k from 3 to 10 improves ASR across most target models, confirming the usefulness of top-k dark knowledge for black-box surrogate distillation. When k becomes too large, such as 15 or 20, the performance slightly drops because low-probability tail classes may introduce noisy supervision. Thus, $k = 10$ provides a good trade-off between informative dark knowledge and noise, and is used as the default setting.

9) Comparison of Mixing Strategies: This subsection examines whether the benefit of RsMix originates from remote-sensing-aware mixing rather than from generic data augmentation. We compare five mixing strategies under an identical distillation schedule and the same MI-FGSM attack backbone: no mixing, MixUp, CutMix, random RsMix, and confusion-matrix-guided RsMix. The results are summarized in Fig. 9.

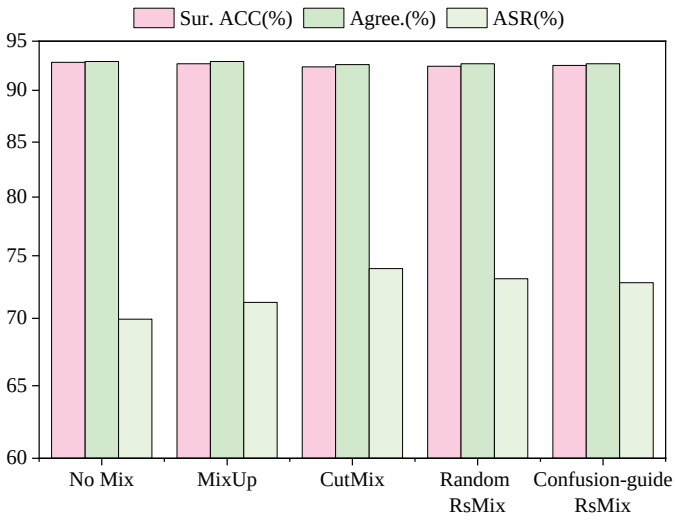

Fig. 9. Ablation study of different sample mixing strategies. Surrogate Acc, Agreement, and ASR are reported in percentage.

Fig. 9 shows that any form of mixed-sample distillation outperforms the no-mixing baseline, raising ASR from 69.93% to above 71%. This validates the central motivation that clean remote sensing samples provide limited dark knowledge, while mixed samples induce richer multi-class responses from the target model. Among all mixing strategies, MixUp yields the smallest performance gain, because global linear interpolation blurs the spatial structures critical for remote sensing scene recognition and weakens discriminative cues learned by the surrogate model. In contrast, spatially localized mixing strategies (CutMix and RsMix) achieve higher ASR. This observation aligns with the inherent characteristics of remote sensing scenes, which are dominated by spatially structured land-cover patterns including runways, ports, residential blocks, and farmland parcels.

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CutMix and RsMix attain comparable ASR under this single-surrogate setup, and RsMix preserves slightly higher target-surrogate agreement than MixUp. The core merit of the confusion-matrix-guided RsMix lies in its explicit bias toward mixing semantically similar category pairs, e.g., dense vs. medium residential areas, ports vs. rivers. These pairs are exactly the classes that the target model frequently confuses. Such semantic-aware mixing improves the information density of each query. We expect this advantage to become more prominent under strict query budgets and for challenging cross-architecture target models, where blind random mixing or simple geometric mixing fails to reliably capture the most informative inter-class relationships.

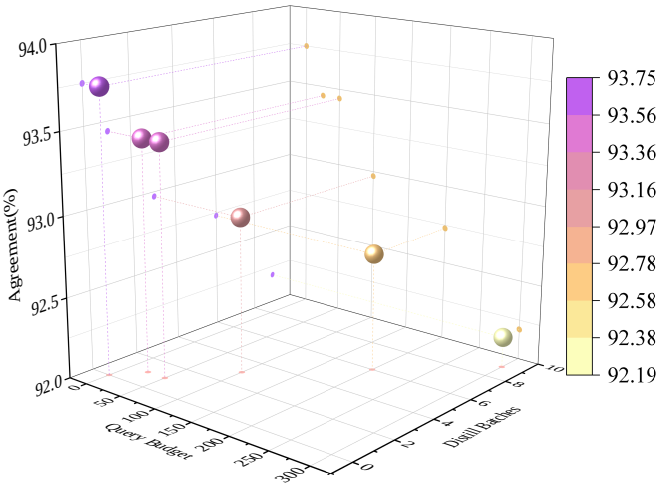

Fig. 10. Agreement results under different query budgets.

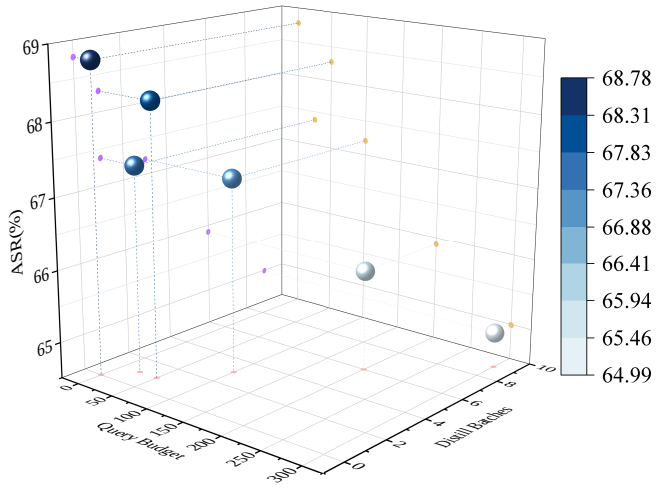

Fig. 11. ASR results under different query budgets.

10) Query Efficiency Evaluation: This subsection evaluates the query efficiency of the proposed method, which is the central concern of a query-efficient black-box attack. We vary the number of target-model queries used for surrogate distillation and measure the resulting Attack Success Rate (ASR). The query budget is approximated by the number of distillation batches, where each batch issues queries for both clean and mixed samples. The quantitative results are summarized in Fig. 10 and Fig. 11.

As shown in Fig. 10 and Fig. 11, the proposed method achieves a competitive ASR of 68.78% even under the zero-query setting. This favorable performance originates from two aspects: the surrogate model inherits transferable adversarial knowledge from the pre-trained source model, and the embedded MI-FGSM module serves as a powerful attack baseline. When equipped with a limited number of target model queries, the ASR remains close to the zero-query baseline, demonstrating that our attack framework does not rely on massive query overhead to preserve satisfactory attack effectiveness. Such property is highly practical for real-world black-box attack scenarios, since each query submitted to the deployed remote sensing target model brings extra computational cost and raises the risk of being detected by defense systems.

We further observe a slight decline in ASR with moderate query budgets in this representative experiment. This phenomenon stems from the short distillation schedule adopted for rapid experimental iteration: with only a small number of distillation batches, the surrogate model weights are disturbed before adequate target domain information is accumulated to stabilize the learned decision boundary. Under the complete experimental pipeline, longer distillation processes with a larger query budget enable the confusion matrix and surrogate model to fully converge. After convergence, the query-distilled surrogate is capable of outperforming the pure zero-query transfer attack baseline. Notably, all queries consumed during surrogate distillation are fully counted into the total query budget. Therefore, the reported query efficiency accurately reflects the real overhead of accessing the black-box target model, without any implicit uncounted query advantage.

11) Dark Knowledge Visualization: This subsection quantitatively visualizes the dark knowledge that inspires our proposed RsMix strategy. We feed clean and mixed remote sensing samples into the black-box target model and record the top-1 probability as well as the top-k entropy derived from the model's softmax outputs. A larger top-k entropy implies that more non-top-1 categories are assigned considerable probability values, which corresponds to more abundant dark knowledge. The results are listed in Fig. 12.

As presented in Fig. 12, clean remote sensing images lead to extremely sharp output distributions of the target model, with an average top-1 probability of 0.9988 and nearly vanishing top-k entropy. Such sharp softmax responses barely contain any inter-class correlation information, which explains why surrogate distillation only using clean samples gains limited performance improvement. By contrast, mixed samples significantly raise the top-k entropy. This observation verifies that sample mixing activates multiple semantically correlated categories in the target model and excavates the implicit inter-class relational structure hidden in model outputs.

Compared with CutMix, the proposed RsMix achieves prominent entropy growth while retaining a higher top-1 probability. This indicates that RsMix enriches dark knowledge without thoroughly destroying the dominant semantic information of original scenes. Such a trade-off matches the core design principle of RsMix: maximizing the dark knowl-

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TABLE X
COMPARISON OF TOP-5 PROBABILITY DISTRIBUTIONS FOR CLEAN AND ADVERSARIAL SAMPLES

Rank	Clean Class	Clean Prob	Adversarial Class	Adversarial Prob
1	Bridge	0.9999	Park	0.99997
2	Port	7.0×10^{-5}	Resort	2.2×10^{-5}
3	Park	1.5×10^{-6}	River	6.3×10^{-7}
4	StorageTanks	5.0×10^{-7}	SparseResidential	4.8×10^{-7}
5	Airport	2.8×10^{-7}	StorageTanks	4.8×10^{-7}

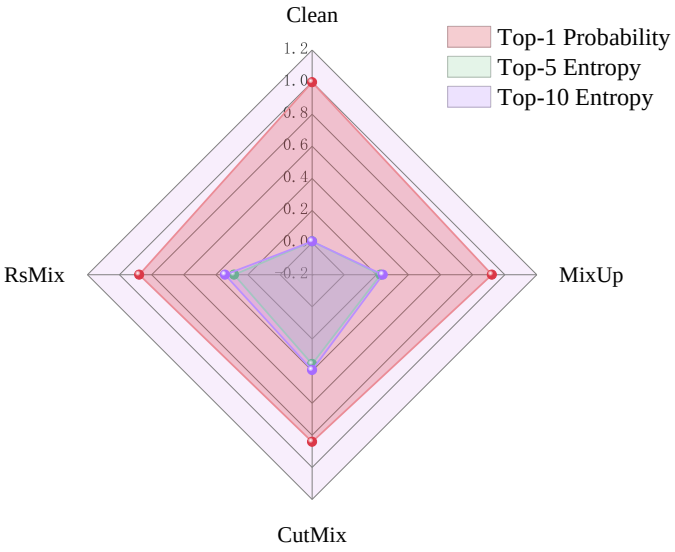

Fig. 12. Top-1 probability and top-k entropy under different sample construction strategies.

edge density of each query sample while constraining mixed samples within the natural manifold of remote sensing data, instead of blindly pursuing high entropy at the expense of semantic rationality.

12) Top-k Probability Shift Analysis: This subsection investigates how the crafted adversarial perturbation distorts the target model's classification decisions from the perspective of top-k category probability distribution. For a typical successfully attacked sample, we compare the top-k probability distribution predicted by the target model before and after adversarial perturbation. Experimental results are summarized in Table X.

As illustrated in Table X, the clean sample is confidently classified as Bridge with probability approaching 1, whereas its adversarial counterpart is predicted as Park with equally high confidence. This phenomenon proves that the attack proposed in this paper does not merely generate minor misclassification near decision boundaries; instead, it pushes the sample far into a completely different decision region of the black-box target model.

The rearrangement of low-rank categories further demonstrates that adversarial perturbation reconstructs the overall local category ranking, rather than only flipping the top-1 prediction label. This characteristic aligns with the optimization objective of our entire framework: we train the surrogate model to fit the dark knowledge structure of the target model, such that adversarial gradients calculated on the surrogate can effectively modify the probability landscape of the black-box target model.

Moreover, the adversarial sample owns a top-1 confidence comparable to that of the clean sample. This manifests that our attack is not only effective but also robust: the target model makes firm false predictions with high certainty.

V. DISCUSSION

Experimental results show that the proposed framework improves most when the surrogate is a normalized network with skip or dense connectivity (ResNet-18, DenseNet-121), and can fall below even simple iterative baselines under shallow or narrow surrogates (AlexNet, RegNetX-400MF), sometimes by a wide margin (e.g., AlexNet→ResNet-101 on AID, where ASR drops to 46.72% against 80.73%). On MLRSNet this dispersion inverts the overall ranking: the framework leads on AID and NWPU-RESISC45 by 3.96 and 12.20 percentage points yet trails DeCoWA by 3.58 points, a deficit contributed entirely by AlexNet (-7.22) and RegNetX-400MF (-6.64) while the two expressive surrogates stay level with the strongest baseline. The reason lies in the top-k dark-knowledge distillation and classification feature alignment modules, which transfer *relational* rather than label-level supervision: the normalized top-k response specifies ratios among semantically adjacent categories, and $\mathcal{L}_{cf}$ requires the penultimate feature to align with a prototype spanning several classifier directions. Batch-normalized multi-path architectures keep logit scales stable under soft-target training and preserve mid-level multi-scale evidence up to the classifier, so this structure can be embedded. AlexNet's stride-4 stem removes the fine texture and layout cues that separate exactly the confusable pairs RsMix selects, while RegNetX-400MF's narrow grouped trunk yields a penultimate space with a rapidly decaying spectrum, in which the prototype collapses onto the top-1 direction and $\mathcal{L}_{cf}$ degenerates towards hard-label supervision. Both surrogates receive dark knowledge they cannot retain, and the effect compounds with the label space, which is why they degrade furthest on the 46-class benchmark.

A second, independent condition is gradient compatibility, which distillation cannot alter because it is fixed by architecture rather than learned from soft labels. Across all three datasets the best target for AlexNet is VGG-19 (67.63%, 77.76%, 79.18%) and the best target for RegNetX-400MF is RegNetX-8GF (66.65%, 62.73%, 69.39%), each pairing a

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surrogate with the target sharing its design principle rather than its capacity. The same mechanism explains the target side: ViT-B and Swin-T average out localized perturbation energy through global or windowed attention, whereas MobileViT-S retains a convolutional stem and is attacked far more easily.

Together these conditions explain why a mismatched surrogate underperforms rather than merely matches the baselines. The framework already reaches 68.78% ASR at zero queries, since the undistilled source carries a strong generic transfer prior; distillation deliberately trades that prior for a target-specific one. Where both conditions hold the exchange is clearly profitable (DenseNet-121 against ResNet-50: 82.54% on AID and 81.34% on MLRSNet, versus 67.13% and 78.54% for the best baseline); where they do not, the surrogate abandons the generic prior for a corrupted approximation of the target and serves neither role. Higher variance across surrogates is thus intrinsic to target-specific reshaping, and the surrogate should be selected rather than taken as given: a moderately deep, batch-normalized backbone is preferable to a larger but structurally alien one, as the RegNetX-400MF→RegNetX-8GF pairing shows. The framework nonetheless inherits the inductive bias of the single surrogate it distils, which bounds its effectiveness on pure attention targets and leaves isolated failures such as ResNet-101 on MLRSNet unexplained by either condition; distilling an architecturally diverse ensemble is a natural way to remove this dependence. Overall, the four modules act in concert to raise the information density of each query and reshape the surrogate towards the victim, yielding strong transferability whenever the surrogate can absorb the target's class-relation structure and shares its gradient geometry.

VI. Conclusion

This article proposed a dark-knowledge-guided surrogate distillation framework for query-efficient black-box adversarial attacks on remote sensing image classification. The proposed method uses RsMix to construct high-dark-knowledge mixed samples and queries the target model to obtain informative top-k responses. These responses are used to update the category confusion matrix and guide surrogate distillation. To improve surrogate fidelity, a top-k dark-knowledge distillation loss and a classification feature alignment loss are introduced. During adversarial generation, a manifold consistency constraint is further used to preserve remote sensing semantic structure under the L_∞ perturbation budget. Experiments on AID, NWPU-RESISC45 and MLRSNet demonstrate that the proposed method achieves effective transferability across CNN, transformer, and hybrid target models. Visualization results also show that the generated adversarial examples shift model attention from discriminative regions to irrelevant areas, leading to misclassification while maintaining visual similarity.

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